Robust local efficiency of an equilibrated-flux a posteriori error estimator for algebraically stabilised convection–diffusion–reaction problems

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Abstract

We analyse an equilibrated-flux a posteriori error estimator for conforming P_1 approximations of convection–diffusion–reaction problems with algebraic stabilisation, such as algebraic flux correction. The estimator extends the construction of Smears and Vohralík from reaction–diffusion to convection, and treats the stabilisation as an additional cellwise diffusion tensor. It gives a guaranteed upper bound on the augmented energy norm of the error plus the dual norm of the stabilisation functional. The bound holds for every non-negative reaction and for every conforming discrete function, so it applies to inexact iterates of the nonlinear solver. We prove that the estimator is locally efficient with respect to the dual norms of the residual functionals. The local constants depend only on the space dimension and the shape regularity of the mesh; the global constant may in addition depend on the stability constant of a commuting projection. None of them depends on the diffusion, convection and reaction coefficients, the mesh size or the stabilisation. The local flux problems carry no weights; robustness in the reaction-dominated regime comes from the way the fluxes are evaluated, together with a best-approximation argument. In terms of the error, the bound is two-sided with a factor $(1 + c_b)^2$, where c_b measures the variation of the effective reaction, and with a factor $1 + c_b$ in a weighted error measure. We also prove three further results. First, the estimator is locally efficient in a localisable error measure whose only non-local term is the dual norm of the nodal moments of the stabilisation, a quantity the estimator computes. Second, a stopping criterion for the nonlinear iteration is reliable and efficient at every iterate, without any smallness condition. Third, the analysis extends to variable diffusion at the price of an $O(h^2)$ oscillation term. Numerical experiments in two and three space dimensions illustrate the results.

Keywords: a posteriori error estimation; equilibrated fluxes; convection–diffusion–reaction equations; algebraic flux correction; robustness; adaptive stopping criteria.
MSC (2020): 65N15, 65N30, 65N50.

1 Introduction

We consider the convection–diffusion–reaction problem

$$-\varepsilon\Delta u + b\cdot\nabla u + cu = f \text{ in } \Omega, \qquad u = u_D \text{ on } \Gamma_D, \qquad \varepsilon\nabla u\cdot n = g_N \text{ on } \Gamma_N, \tag{1.1}$$

in the convection-dominated regime $\varepsilon \ll |b|$, discretised by conforming P_1 finite elements with an algebraic stabilisation. Algebraic stabilisations, such as algebraic flux correction (AFC) [BJK16, BJK17], the monotone upwind-type algebraic stabilisations [Kn23] and edge-based nonlinear diffusion [BBK17], add a nonlinear, solution-dependent artificial diffusion. They satisfy the discrete

maximum principle and avoid the spurious oscillations of linear stabilisations. The resulting nonlinear systems are solved iteratively, often with many iterations, and the iteration is usually stopped at a fixed algebraic tolerance.

For adaptivity, and to stop the iteration, one needs an a posteriori error estimator with two properties. It should be *guaranteed*: a computable upper bound of the error, without unknown constants. And it should be *robust*: its effectivity should not depend on ε, b or c.

State of the art.

- **Algebraic stabilisations.** Allendes, Barrenechea and Rankin [ABR17] give a guaranteed bound for AFC-type schemes in the energy norm. Its local lower bound degenerates with the local Péclet number. Residual-based estimators for AFC and algebraic stabilisations [J21, J24, AJ27] are not guaranteed and are not robust.
- **Linear stabilisations.** Verfürth [V05] and Tobiska and Verfürth [TV15] prove robustness in the energy norm augmented by a dual norm of the convective derivative, with unknown constants. Eigel and Merdon [EM16] give guaranteed bounds in this augmented norm. Cai and Cai [CC25] prove robust global efficiency of hybrid estimators.
- **Discontinuous Galerkin.** Ern, Stephansen and Vohralík [ESV10] give guaranteed and robust bounds. Their local lower bounds are only semi-robust, with cut-offs in the local Péclet and Damköhler numbers.
- **Reaction-dominated regime.** Equilibrated-flux estimates that are robust for reaction–diffusion need either a combination with a residual-type estimate [CFPV09] or weights in the flux construction [SV20]. Ainsworth and Vejchodský [AV11, AV14] obtain fully computable robust bounds, for constant and piecewise-constant reaction coefficients.
- **Localisation and stopping.** Dual norms of functionals that vanish on a partition of unity localise over vertex patches [BMV20, CV18]. Verfürth [V05] and Cai and Cai [CC25] prove efficiency only globally in the augmented norm; Sangalli [Sa08] uses a different, almost robust norm in one dimension; space-time dual norms are used for unsteady advection–diffusion in [DEV13]; for local discontinuous Galerkin methods the augmented norm has been used with a reliable estimator whose efficiency was left open [LDG20]. A posteriori stopping criteria for iterative solvers go back to [EV13], with multilevel algebraic fluxes in [PRVW20] and quasi-optimal cost for strongly monotone operators in [HPSV21].
- **Variable coefficients.** Ainsworth and Vejchodský [AV14] are robust in the size of a piecewise-constant reaction, not in a diffusion contrast; Ern, Stephansen and Vohralík [ESV10] allow inhomogeneous diffusion with constants depending on the local contrast.

To our knowledge, no estimator for nonlinear algebraic stabilisations is both guaranteed and robust, with a proof of efficiency; nor has the augmented norm been localised for stabilised conforming schemes, with the nodal moments of the stabilisation as the only global term (Theorem 7.2). We are not aware of an a posteriori stopping criterion for algebraic flux correction. Theorem 9.2 is an adaptation of known arguments.

Contributions. We analyse the estimator obtained by extending the construction of Smears and Vohralík [SV20] to convection. The stabilisation enters through the diffusion tensor $D = \varepsilon I + A$, where A is the cellwise artificial diffusion of the scheme. The error measure is

$$E = |||e||| + \|B'_A(e,\cdot)\|_* + \|\tilde{S}\|_*,$$

the augmented energy norm of the error plus the dual norm of the stabilisation functional (Section 2). Our results are as follows.

1. **Reliability** (Theorem 4.1). $E \leq \eta_E$ for every $\sigma \geq 0$ and every conforming u_h, with fully computable constants. The constant in the augmented-norm bound is $\kappa = 2 + \max\{1, c_b\}$ (Lemma 2.3); it reduces to the value 3 of [EM16] when the effective reaction $c - \frac{1}{2}\operatorname{div} b$ is constant.

2. **Robust local efficiency** (Theorem 4.2). For each of the two functionals that the estimator equilibrates, the local contribution is bounded by the local dual norm of the functional on a vertex neighbourhood, plus data oscillation. The local constants depend only on d and the shape regularity, the global one also on the constant of a commuting projection; none depends on ε, σ, b, c, h, the stabilisation or u_h.

3. **A weight-free construction.** The local flux problems carry no weights; the weights enter only through the evaluation (Lemma 5.6). The reaction-dominated case is handled by a comparison with the best approximation (Lemma 5.9). This is consistent with the necessity of weights shown in [SV20], because the evaluation is a combination of two estimators (Remark 5.7).

4. **Two-sided bounds in the error** (Corollaries 4.3 and 4.4). $E \leq \eta_E \leq C(1 + c_b)^2(E + \operatorname{osc})$, and the factor becomes $1 + c_b$ in a weighted measure E_c.

5. **Localisable error measure** (Theorem 7.2). In a measure that is a sum of local contributions, the estimator is locally efficient. The only non-local part of E is the dual norm of the nodal moments of the stabilisation, which the estimator computes.

6. **Adaptive stopping** (Proposition 8.1). Splitting the estimator into a discretisation and an algebraic part gives a stopping criterion that is reliable and efficient at every iterate of every solver, for every value of the stopping parameter.

7. **Variable diffusion** (Theorem 9.2). With the cellwise L^2 projection of ε onto P_1 in the fluxes, reliability holds with an oscillation term of order h^2, and local efficiency holds with constants depending on the local contrast of ε.

Outline. Section 2 states the setting and the error measures, and Section 3 the estimator. The main results are stated in Section 4. Section 5 collects the tools and proves the key lemmas, and Section 6 proves the main results. Sections 7, 8 and 9 treat the localisable error measure, adaptive stopping and variable diffusion. Section 10 reports numerical experiments, and Section 11 concludes with open problems.

2 Setting

2.1 Hypotheses

(H1) $\Omega \subset \mathbb{R}^d$ $(d = 2, 3)$ is a connected polytope, and $\mathcal{T}_h$ is a conforming simplicial mesh with shape-regularity parameter ϑ. The boundary partition is assumed to be mesh-aligned: Γ_D and Γ_N are unions of boundary facets.

(H2) $\varepsilon > 0$ is constant (variable ε is treated in Section 9); $b \in W^{1,\infty}(\Omega)^d$, $c \in L^\infty(\Omega)$, $f \in L^2(\Omega)$ and $g_N \in L^2(\Gamma_N)$.

(H3) $c - \frac{1}{2}\operatorname{div} b \geq \sigma \geq 0$, and

$$c_b := \min\Big\{ \frac{\|c - \frac{1}{2}\operatorname{div} b\|_\infty}{\sigma},\ \frac{\|c - \frac{1}{2}\operatorname{div} b\|_\infty C_F^2}{\varepsilon} \Big\} < \infty,$$

where C_F is the Friedrichs constant of $H^1_D(\Omega)$, and the first entry is ∞ if $\sigma = 0$. For a computationally certified bound, σ and an upper bound for $\|c - \frac{1}{2}\operatorname{div} b\|_\infty$ are assumed to be supplied as certified input bounds; sampled coefficient values alone do not establish (H3). In computing c_b, this upper bound may replace the exact essential supremum, giving a possibly larger but still valid constant.

(H4) Γ_D has positive measure and contains the inflow boundary, so that $b \cdot n \geq 0$ on Γ_N.

(H5) u_D is the trace of a P_1 function.

(H6) $u_h \in V_h = P_1(\mathcal{T}_h) \cap H^1(\Omega)$ with $u_h = u_D$ on Γ_D. The stabilisation at this iterate admits a representation $S(u_h; u_h, v_h) = (A_h \nabla u_h, \nabla v_h)$ for all $v_h \in V_{h,0} = V_h \cap H^1_D(\Omega)$, where A_h is cellwise constant, symmetric and positive semidefinite, and may depend on u_h. Nothing is assumed about how u_h was obtained; it may be any iterate of the nonlinear solver.

Here $H^1_D(\Omega) = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}$. Hypothesis (H3) is used only for the conversion between dual norms of the residual and the error (Lemma 2.3); the efficiency result, Theorem 4.2, does not use it.

One further assumption is needed, and only for the global efficiency bound.

Assumption 2.1 (commuting projection [CW08, Licht19])**.** *There is a projection Π_h from $\{s \in H(\operatorname{div})(\Omega) : s \cdot n = 0 \text{ on } \Gamma_N\}$ onto the RT_0 fields with the same boundary condition, such that $\|\Pi_h s\| \leq C_\Pi \|s\|$ and $\operatorname{div} \Pi_h s = \Pi_0 \operatorname{div} s$. For mixed boundary conditions, C_Π is required to be uniform over the shape-regular mesh family on the fixed pair (Ω, Γ_N); no uniformity over varying domains or boundary partitions is claimed.*

For $\Gamma_N = \emptyset$ the smoothed projection of Christiansen and Winther [CW08] has these properties; for mixed boundary conditions, Licht [Licht19] constructs commuting smoothed projections that preserve homogeneous boundary conditions on a part of the boundary. The assumption is used only for the global efficiency bound of the defect term (Lemma 5.10(ii), Remark 6.2).

Remark 2.2 (edge-difference stabilisations satisfy (H6)). Let $\mathcal{E}_h$ be the set of unoriented mesh edges, and suppose the stabilisation has the edge-difference form

$$S(w; u, v) = \sum_{E=\{i,j\} \in \mathcal{E}_h} \beta_E(w)(u(x_j) - u(x_i))(v(x_j) - v(x_i)), \qquad \beta_E(w) \geq 0,$$

where the coefficient is attached to the unordered edge (so it is unchanged if i and j are interchanged). For each edge, let n_E be the number of cells containing it, choose either orientation and set $t_E = x_j - x_i$. Define the cellwise tensor

$$A(w)|_K = \frac{1}{|K|} \sum_{E \subset K} \frac{\beta_E(w)}{n_E}\, t_E \otimes t_E.$$

For any conforming piecewise-affine u, v, the constant gradients on each cell satisfy $u(x_j) - u(x_i) = \nabla u|_K \cdot t_E$ for every $K \supset E$, and likewise for v. Consequently,

$$\begin{aligned}(A(w)\nabla u, \nabla v) &= \sum_{K \in \mathcal{T}_h} \sum_{E \subset K} \frac{\beta_E(w)}{n_E} (\nabla u|_K \cdot t_E)(\nabla v|_K \cdot t_E) \\ &= \sum_{E \in \mathcal{E}_h} \beta_E(w)(u(x_j) - u(x_i))(v(x_j) - v(x_i)) = S(w; u, v).\end{aligned}$$

Thus $A(w)|_K$ is symmetric positive semidefinite, and setting $w = u = u_h$ proves (H6) with $A_h = A(u_h)$. This is a sufficient condition, not a claim about every algebraic stabilisation: non-symmetric/directed limiters and forms not expressible as nonnegative edge-difference sums are not covered. In particular, symmetric-limiter AFC variants have this form with $\beta_E = -(1-\alpha_E)d_E$ when $d_E \le 0$ and $0 \le \alpha_E \le 1$; the symmetrised method of [Kn23] and the edge-based nonlinear diffusion of [BBK17] are further examples.

2.2 Norms, weights and functionals

Norms. For $\omega \subseteq \Omega$, let

$$|||v|||_\omega^2 = \varepsilon\|\nabla v\|_\omega^2 + \sigma\|v\|_\omega^2, \qquad H^1_*(\omega) = \{v \in H^1(\omega) : v = 0 \text{ on } \partial\omega \setminus \Gamma_N\},$$
$$\|\ell\|_{*,\omega} = \sup_{v \in H^1_*(\omega)\setminus\{0\}} \frac{\ell(v)}{|||v|||_\omega}, \qquad \|\ell\|_* = \sup_{v \in H^1_D(\Omega)\setminus\{0\}} \frac{\ell(v)}{|||v|||}.$$

Extension by zero maps $H^1_*(\omega)$ into $H^1_D(\Omega)$, so $\|\ell\|_{*,\omega}$ is monotone in ω and $\|\ell\|_{*,\omega} \le \|\ell\|_*$. Test functions in $H^1_*(\omega)$ need not vanish on Γ_N, which is needed for hat functions and face bubbles at Neumann vertices and faces (Lemmas 5.10 and 5.11).

Weights. For an element or a face S with diameter h_S,

$$\alpha_S = \min\{h_S\varepsilon^{-1/2},\, \sigma^{-1/2}\}, \qquad \theta_S = \frac{\varepsilon^{1/2}\sigma^{-1/2}}{h_S}, \qquad \text{so that} \qquad \alpha_S = h_S\varepsilon^{-1/2}\min\{1, \theta_S\},$$

with $\alpha_S = h_S\varepsilon^{-1/2}$ and $\theta_S = \infty$ if $\sigma = 0$. The regime $\theta_S \ge 1$ is diffusion-dominated at the mesh scale, and $\theta_S < 1$ reaction-dominated.

Residual functionals. Let $B(w,v) = \varepsilon(\nabla w, \nabla v) + (b\cdot\nabla w + cw, v)$ and $e = u - u_h$. For $v \in H^1_D(\Omega)$,

$$\begin{aligned} \mathrm{Res}(v) &= (f,v) + (g_N, v)_{\Gamma_N} - B(u_h, v) = B(e, v), \\ \tilde{S}(v) &= (A\nabla u_h, \nabla v), \\ \mathrm{Res}_A(v) &= \mathrm{Res}(v) - \tilde{S}(v) = (r, v) - ((\varepsilon I + A)\nabla u_h, \nabla v) + (g_N, v)_{\Gamma_N}, \end{aligned}$$

with $r = f - b\cdot\nabla u_h - c\,u_h$. Both Res_A and $-\tilde{S}$ have the generic form

$$\ell(v) = (r_\ell, v) - (D\nabla u_h, \nabla v) + (g_\ell, v)_{\Gamma_N}, \tag{2.1}$$

with $(r_\ell, D, g_\ell) = (r, \varepsilon I + A, g_N)$ for Res_A and $(0, A, 0)$ for $-\tilde{S}$.

Error measures. Write $B = B'_S + B'_A$, an exact splitting, with B'_S symmetric and B'_A skew-symmetric up to a boundary term, $B'_A(w,v) + B'_A(v,w) = \int_{\Gamma_N} b\cdot n\, wv$ for $w, v \in H^1_D(\Omega)$:

$$B'_A(w,v) = (b\cdot\nabla w + \tfrac{1}{2}(\operatorname{div} b)\, w,\ v), \qquad B'_S(w,v) = \varepsilon(\nabla w, \nabla v) + ((c - \tfrac{1}{2}\operatorname{div} b)\, w,\ v).$$

The augmented norm and the error measure are

$$|||e|||_\oplus = |||e||| + \|B'_A(e,\cdot)\|_*, \qquad E = |||e|||_\oplus + \|\tilde{S}\|_*.$$

The term $\|\tilde{S}\|_*$ measures the consistency error of the stabilisation. It is the quantity through which the stabilisation enters the residual.

Lemma 2.3 (augmented norm and residual)**.** *Assume (H1)–(H5) and $e \in H^1_D(\Omega)$. Then*

$$|||e||| \le \|\operatorname{Res}\|_*, \qquad |||e|||_\oplus \le \kappa \|\operatorname{Res}\|_*, \qquad \|\operatorname{Res}\|_* \le \max\{1, c_b\}\, |||e|||_\oplus,$$

with $\kappa = 2 + \max\{1, c_b\}$.

Proof. Integrating by parts, $B(v,v) = \varepsilon\|\nabla v\|^2 + ((c - \frac{1}{2}\operatorname{div} b)v, v) + \frac{1}{2}\int_{\Gamma_N} b\cdot n\, v^2 \ge |||v|||^2$ for $v \in H^1_D(\Omega)$, by (H3) and (H4). With $v = e$ this gives $|||e|||^2 \le \operatorname{Res}(e) \le \|\operatorname{Res}\|_* |||e|||$. Next,

$$|B'_S(w,v)| \le \varepsilon^{1/2}\|\nabla w\|\, \varepsilon^{1/2}\|\nabla v\| + \|c - \tfrac{1}{2}\operatorname{div} b\|_\infty \|w\|\|v\| \le \max\{1, c_b\}|||w|||\,|||v|||,$$

using $\|w\|\|v\| \le \sigma^{-1}|||w|||\,|||v|||$ for the first entry of c_b, and $\|w\|\|v\| \le C_F^2\varepsilon^{-1}|||w|||\,|||v|||$ (Friedrichs inequality) for the second. Since $B'_A(e,v) = \operatorname{Res}(v) - B'_S(e,v)$, we get $\|B'_A(e,\cdot)\|_* \le \|\operatorname{Res}\|_* + \max\{1, c_b\}|||e||| \le (1 + \max\{1, c_b\})\|\operatorname{Res}\|_*$, and the second claim follows. The third follows from the same splitting. □

Remark 2.4 (computing c_b). The value $\kappa = 3$ of [EM16] corresponds to $c - \frac{1}{2}\operatorname{div} b \equiv \sigma$, so $c_b = 1$. For a variable reaction with a small lower bound σ, the Friedrichs entry of c_b can be much smaller than the σ-based entry, and it is the only one available for $\sigma = 0$. A guaranteed C_F is available in two cases. Without Neumann boundary, the H^1_0 constant of a bounding box can be used, by extension by zero. On a box $\prod_i (0, L_i)$ with Neumann faces, separation of variables gives $C_F^{-2} = \sum_i \lambda_i$, where $\lambda_i = (\pi/L_i)^2$ if both faces normal to e_i are Dirichlet faces, $(\pi/2L_i)^2$ if exactly one is, and 0 otherwise; a partly Neumann face is counted as a Neumann face, which only lowers λ_i. For other domains with a Neumann boundary, we keep the σ-based c_b when $\sigma > 0$.

The supremum $\|c - \frac{1}{2}\operatorname{div} b\|_\infty$ must be bounded from above, and the lower bound in (H3) must be verified over each cell; finite sampling alone does not certify either condition for general $c \in L^\infty$ and $b \in W^{1,\infty}$. If $g = c - \frac{1}{2}\operatorname{div} b$ is known to be Lipschitz, an upper bound on $\|\nabla g\|_{L^\infty(K)}$ together with a covering-radius bound for the sample set gives a certified enclosure. The implementation's sampled check is therefore diagnostic unless those derivative and coverage bounds are themselves certified.

3 The estimator

Let $\mathcal{V}_h$ be the set of vertices. For $a \in \mathcal{V}_h$, let ψ_a be the hat function, $\mathcal{T}_a$ the cells containing a, ω_a their union, $\mathcal{F}_a$ the faces containing a, and h_a the diameter of ω_a. For $a \in \partial\Omega$, the faces of $\partial\omega_a$ that lie on Γ_D are called *free*. A vertex is *compatible* if its patch has no free face; this excludes the vertices on $\overline{\Gamma_D}$ and the Neumann vertices whose patch has a face on Γ_D.

3.1 Local data

For each vertex a and each functional ℓ of the form (2.1), define:

- the flux space $V^a \subset \mathrm{RTN}_1(\mathcal{T}_a) \cap H(\mathrm{div})(\omega_a)$ of the fields v with $v\cdot n = -\Pi_F(\psi_a g_\ell)$ on the faces $F \subset \partial\omega_a \cap \Gamma_N$, no condition on free faces, and $v\cdot n = 0$ on the rest of $\partial\omega_a$;
- the source $g_a = \Pi_1(\psi_a r_\ell - \nabla\psi_a \cdot D\nabla u_h)$, with Π_1 the cellwise L^2 projection onto P_1;
- the *defect* $\delta_a = \ell(\psi_a)$ if a is compatible and $\delta_a = 0$ otherwise, and the shifted source $\tilde{g}_a = g_a - c_a$ with $c_a = \delta_a/|\omega_a|$.

The shift makes $(\tilde{g}_a, 1)_{\omega_a}$ consistent with the prescribed normal fluxes on compatible patches. For $\ell = \mathrm{Res}_A$ at the exact discrete solution, δ_a vanishes up to the quadrature error discussed next. For $\ell = -\tilde{S}$, $\delta_a = -S(u_h; u_h, \psi_a)$ are the nodal moments of the stabilisation, which do not vanish.

Remark 3.1 (inexact quadrature). For $\ell = \mathrm{Res}_A$, the estimator computes δ_a with its own quadrature rule, while the discrete system is assembled with another one. We write $\delta_a = \delta_a^{\mathrm{alg}} + \delta_a^{\mathrm{quad}}$:

- $\delta_a^{\mathrm{alg}} = -r_a$, where r_a is the component of the assembled algebraic residual at a; it vanishes at the discrete solution;
- $\delta_a^{\mathrm{quad}} = \mathrm{Res}_A^{\mathrm{est}}(\psi_a) - \mathrm{Res}_A^{\mathrm{asm}}(\psi_a)$ is the difference of the two rules on all data terms tested with ψ_a: f, $b \cdot \nabla u_h$, $c\, u_h$, the diffusion term and g_N. The stabilisation term is integrated exactly.

δ_a^{quad} vanishes when the assembly integrates every data term exactly; with P_1 test functions and an assembly rule of degree 4, as in our implementation, this holds for ε of degree at most 4, f and b of degree at most 3, and c of degree at most 2. Otherwise it is of the order of the quadrature error. We write $\psi^{\mathrm{quad}} = \sum_a \delta_a^{\mathrm{quad}} \chi_{\omega_a}/|\omega_a|$ and $\mathrm{osc}_q := \|\psi^{\mathrm{quad}}\|_*$. Reliability is not affected, because the full defect δ_a is shifted and bounded.

3.2 Patch problems

For $\sigma > 0$, let

$$\sigma^a = \operatorname*{argmin}_{v \in V^a} J_a(v), \qquad J_a(v) = \varepsilon^{-1} \|\psi_a D\nabla u_h + v\|_{\omega_a}^2 + \sigma^{-1} \|\operatorname{div} v - \tilde{g}_a\|_{\omega_a}^2.$$

For $\sigma = 0$, the flux term is minimised subject to $\operatorname{div} v = \tilde{g}_a$, which is solvable on compatible patches because of the shift. We write $\tau^a = \psi_a D\nabla u_h + \sigma^a$ and $m^a = \operatorname{div} \sigma^a - \tilde{g}_a$, with $m^a = 0$ if $\sigma = 0$, and set

$$\sigma_h = \sum_a \sigma^a, \qquad \tau = D\nabla u_h + \sigma_h = \sum_a \tau^a \text{ on each cell.}$$

The patch problems have no weights; this is the difference from [SV20]. In every estimator formula below, terms weighted by σ^{-1} or $\sigma^{-1/2}$ are omitted when $\sigma = 0$, and the associated minimisation problems impose exact equilibration. In minimum formulas, the reaction-scale entry is interpreted as $+\infty$.

3.3 Evaluation

On each cell K, the indicator is

$$\eta_{\ell,K} = \min\Big\{\varepsilon^{-1/2}\|\tau\|_K,\ \sum_{F \subset \partial K} c_F \|\tau \cdot n\|_F + m_K \|\operatorname{div} \tau\|_K\Big\} + \sigma^{-1/2}\Big\|\sum_a m^a\Big\|_K + m_K \|r_\ell - \Pi_1 r_\ell\|_K, \tag{3.1}$$

where

$$m_K = \min\Big\{\frac{h_K}{\pi \varepsilon^{1/2}},\ \sigma^{-1/2}\Big\}, \qquad c_F^2 = \frac{|F|}{|K|} \min\Big\{\frac{h_K^2}{\varepsilon}\Big(\frac{1}{\pi^2} + \frac{2}{d\pi}\Big),\ \frac{1}{\sigma} + \frac{h_K}{d(\sigma\varepsilon)^{1/2}}\Big\}.$$

The defects of the compatible patches are collected in $\psi_{\mathrm{def}} = \sum_a c_a \chi_{\omega_a}$ and bounded by one global lowest-order problem,

$$\eta_{\ell,\mathrm{def}} = \min_{\substack{s \in \mathrm{RT}_0(\mathcal{T}_h) \\ s \cdot n = 0 \text{ on } \Gamma_N}} \big(\varepsilon^{-1}\|s\|^2 + \sigma^{-1}\|\psi_{\mathrm{def}} - \operatorname{div} s\|^2\big)^{1/2},$$

with the constraint $\operatorname{div} s = \psi_{\mathrm{def}}$ for $\sigma = 0$. The estimator for ℓ is

$$\eta_\ell = \Big(\sum_K \eta_{\ell,K}^2\Big)^{1/2} + \eta_{\ell,\mathrm{def}}.$$

Let η_N be a computable bound of the Neumann data oscillation $\sup_v (g_N - \Pi_F g_N, v)_{\Gamma_N} / |||v|||$, obtained from the trace inequality. We set

$$\eta_A = \eta_{\mathrm{Res}_A}, \qquad \eta_S = \min\{\eta_S^{\mathrm{alt}},\ \eta_{-\tilde{S}}\}, \qquad \eta = \eta_A + \eta_N + \eta_S, \qquad \eta_E = \kappa\,\eta + \eta_S,$$

where η_S^{alt} is any other guaranteed bound of $\|\tilde{S}\|_*$ (in our implementation, the one built on the fluxes of [ABR17]). Taking the minimum preserves both reliability and efficiency.

Remark 3.2 (the direct variant). The *direct* variant equilibrates Res itself, with $D = \varepsilon I$, and requires $\sigma > 0$. It is not compatible in general, since $\mathrm{Res}(\psi_a) = S(u_h; u_h, \psi_a)$ at the discrete solution, so no efficiency is claimed for it. The implementation replaces $\eta_A + \eta_S$ by $\min\{\eta_{\mathrm{dir}}, \eta_A + \eta_S\}$; this preserves reliability and only lowers the estimator, so the efficiency results below remain valid.

4 Main results

Theorem 4.1 (reliability). *Under (H1)–(H6),*

$$\|\,\mathrm{Res}_A\,\|_* \le \eta_A + \eta_N, \qquad \|\tilde{S}\|_* \le \eta_S, \qquad |||e||| \le \|\,\mathrm{Res}\,\|_* \le \eta, \qquad |||e|||_\oplus \le \kappa\eta, \qquad E \le \eta_E.$$

The oscillation terms of the efficiency result are local. Let $\tilde{\omega}_K$ be the union of the patches ω_a, $a \in \mathcal{V}_K$, and of their neighbours, and

$$\mathrm{osc}_K(\ell) = \alpha_K \|r_\ell - \Pi_1 r_\ell\|_K + \sum_{F \subset \partial K \cap \Gamma_N} \varepsilon^{-1/4} \alpha_F^{1/2} \|g_\ell - \Pi_F g_\ell\|_F, \qquad \mathrm{osc}(\ell)^2 = \sum_K \mathrm{osc}_K(\ell)^2.$$

Theorem 4.2 (robust efficiency). *Assume (H1), (H2), (H4)–(H6) and Assumption 2.1, and let* $\ell \in \{\mathrm{Res}_A, -\tilde{S}\}$. *Then, for every cell* K,

$$\eta_{\ell,K}^2 \le C_{\mathrm{loc}} \sum_{K' \subset \tilde{\omega}_K} \big[\|\ell\|_{*,\tilde{\omega}_{K'}}^2 + \mathrm{osc}_{K'}(\ell)^2\big],$$

and globally

$$\eta_\ell \le C_{\mathrm{glob}} \big(\|\ell\|_* + \mathrm{osc}(\ell)\big).$$

C_{loc} depends only on d and ϑ, and the local bound does not need Assumption 2.1. The global constant C_{glob} may additionally depend on C_Π and hence on the fixed domain and boundary partition, but is independent of ε, σ, b, c, h, A and u_h.

Corollary 4.3 (robust two-sided bound). *Under (H1)–(H6) and Assumption 2.1,*

$$E \le \eta_E \le C(1 + c_b)^2 (E + \mathrm{osc}),$$

with $\mathrm{osc} = \mathrm{osc}(\mathrm{Res}_A) + \mathrm{osc}(-\tilde{S}) + \eta_N$ *and* C *as in Theorem 4.2.*

The factor $(1+c_b)^2$ is the product of κ in the upper bound and of $\max\{1, c_b\}$ in the lower bound of Lemma 2.3. In a measure that weights the energy norm by $M = \max\{1, c_b\}$ it becomes linear.

Corollary 4.4 (weighted measure). *Let* $M = \max\{1, c_b\}$ *and*

$$|||e|||_{\oplus,c} = M|||e||| + \|B_A'(e,\cdot)\|_*, \qquad E_c = |||e|||_{\oplus,c} + \|\tilde{S}\|_* = E + (M-1)|||e|||, \qquad \eta_{E,c} = (M-1)\eta + \eta_E.$$

Under the assumptions of Corollary 4.3, $E_c \le \eta_{E,c} \le C(1 + c_b)(E_c + \mathrm{osc})$.

Remark 4.5. E_c is equivalent to $\|\,\mathrm{Res}\,\|_* + \|\tilde{S}\|_*$ with the constants 1 and $2M+1$. It is the natural measure when $\sigma \ll \|c - \frac{1}{2}\,\mathrm{div}\, b\|_\infty$. A measure free of c_b would need the energy norm with the full effective reaction $c - \frac{1}{2}\,\mathrm{div}\, b$, of which σ is only a lower bound; see Section 11.

5 Tools and key lemmas

5.1 Tools from the literature

Lemma 5.1 (inverse inequalities [SV20, Lemma 2.2]). *For $\tau \in \mathrm{RTN}_1(K)$, $h_K^{1/2}\|\tau\cdot n\|_{\partial K} \le C_\partial \|\tau\|_K$ and $h_K\|\operatorname{div}\tau\|_K \le C_{\mathrm{div}}\|\tau\|_K$, with constants depending only on d and ϑ.*

Lemma 5.2 (patch stability [SV20, Lemma 4.1], [EV15]). *Let $g \in P_1(\mathcal{T}_a)$ and let τ be a broken RTN_1 field on $\mathcal{T}_a$; on compatible patches assume that $(g,1)_{\omega_a}$ equals the total prescribed normal flux. Then*

$$\min\{\|\tau+v\|_{\omega_a} : v\in V^a,\ \operatorname{div} v = g\} \le C\sup\{(g,w)_{\omega_a} - (\tau,\nabla w)_{\omega_a} :\ w\in H^1_\dagger(\omega_a),\ \|\nabla w\|_{\omega_a}=1\},$$

where $H^1_\dagger(\omega_a)$ is the space of functions with mean zero on ω_a (compatible patches), or vanishing on $\partial\omega_a\cap\Gamma_D$ (patches with a free face). With Neumann faces, the prescribed normal flux is removed by the standard affine shift [EV15].

Lemma 5.3 (squeezed bubbles [V05, Lemma 3.6]). *Let b_K be the element bubble and b_F^θ the face bubble squeezed by $\theta_F = \min\{\varepsilon^{1/2}\alpha_F/h_F, 1\}$. For $w\in P_1(K)$ and $\mu\in P_1(F)$,*

$$(w, b_K w)_K \ge c\|w\|_K^2, \qquad |||b_K w|||_K \le C\alpha_K^{-1}\|w\|_K,$$
$$(\mu, b_F^\theta\mu)_F \ge c\|\mu\|_F^2, \qquad |||b_F^\theta\mu|||_{\omega_F} \le C\varepsilon^{1/4}\alpha_F^{-1/2}\|\mu\|_F, \qquad \|b_F^\theta\mu\|_{\omega_F} \le C\varepsilon^{1/4}\alpha_F^{1/2}\|\mu\|_F.$$

Lemma 5.4 (quasi-interpolant, cf. [V05, Lemma 3.3]). *Let $I_c v = \sum_{a\ \text{compatible}} \bar v_{\omega_a}\psi_a$, where $\bar v_\omega$ is the mean of v over ω. Then $|||I_c v||| \le C_I|||v|||$ for $v\in H^1_D(\Omega)$, with C_I depending only on d and ϑ.*

Proof. The σ-part follows from $|\bar v_{\omega_a}| \le |\omega_a|^{-1/2}\|v\|_{\omega_a}$. For the ε-part, on K,

$$\nabla I_c v = \sum_{a\in\mathcal{V}_K}(\bar v_{\omega_a} - \bar v_K)\nabla\psi_a - \sum_{a'\in\mathcal{V}_K\ \text{not compatible}} \bar v_{\omega_{a'}}\nabla\psi_{a'},$$

using $\sum_{a\in\mathcal{V}_K}\nabla\psi_a = 0$. The first sum is bounded by the Poincaré inequality on the vertex patches of K. Each patch in the second sum has a free face on Γ_D, where $v=0$, so the Friedrichs inequality gives $|\bar v_{\omega_{a'}}| \le C h_{a'}^{1-d/2}\|\nabla v\|_{\omega_{a'}}$. The set of vertices excluded in [V05] consists of the vertices on Γ_D; the additional vertices excluded here, whose patch has a face on Γ_D, are treated by the same argument. □

Lemma 5.5 (finite overlap). *If $\omega_1,\dots,\omega_k\subset\Omega$ cover each point at most N times, then $\sum_j \|\ell\|_{*,\omega_j}^2 \le N\|\ell\|_*^2$.*

Proof. Choose $v_j\in H^1_*(\omega_j)$, extended by zero, with $|||v_j|||_{\omega_j}=1$ and $\ell(v_j)\ge(1-\delta)\|\ell\|_{*,\omega_j}$. Set $t_j=\ell(v_j)$ and $v=\sum_j t_j v_j$. Then $\ell(v)=\sum_j t_j^2$ and $|||v|||^2 \le N\sum_j t_j^2$, so $\|\ell\|_* \ge (\sum_j t_j^2)^{1/2}/\sqrt{N}$. Let $\delta\to 0$. □

5.2 The evaluation carries the weight

Let $\varpi_K = \min\{1,\theta_K^{1/2}\}$ and $\varpi_a = \min\{1,\theta_a^{1/2}\}$, where $\theta_a = \varepsilon^{1/2}\sigma^{-1/2}/h_a$. These are the weights of [SV20].

Lemma 5.6 (weighted evaluation). *For $\tau\in\mathrm{RTN}_1(K)$,*

$$\min\Big\{\varepsilon^{-1/2}\|\tau\|_K,\ \sum_{F\subset\partial K} c_F\|\tau\cdot n\|_F + m_K\|\operatorname{div}\tau\|_K\Big\} \le C_A\,\varepsilon^{-1/2}\varpi_K\|\tau\|_K,$$

with C_A depending only on d and ϑ.

Proof. If $\theta_K \geq 1$, the first entry of the minimum gives the bound with $C_A = 1$. Let $\theta_K < 1$. Then $m_K \leq \sigma^{-1/2} = (h_K/\varepsilon^{1/2})\theta_K$. Further, $\frac{1}{\sigma} + \frac{h_K}{d(\sigma\varepsilon)^{1/2}} = \frac{h_K^2}{\varepsilon}(\theta_K^2 + \theta_K/d) \leq 2\frac{h_K^2}{\varepsilon}\theta_K$ and $|F|/|K| \leq C_\vartheta h_K^{-1}$, so $c_F \leq C h_K^{1/2}\varepsilon^{-1/2}\theta_K^{1/2}$. With Lemma 5.1,

$$\sum_F c_F\|\tau\cdot n\|_F + m_K\|\operatorname{div}\tau\|_K \leq C\varepsilon^{-1/2}(\theta_K^{1/2}C_\partial + \theta_K C_{\mathrm{div}})\|\tau\|_K \leq C_A\varepsilon^{-1/2}\theta_K^{1/2}\|\tau\|_K. \qquad \square$$

Remark 5.7 (consistency with the necessity of weights). Smears and Vohralík [SV20] show that weights in the flux construction are necessary for the robustness of any flux-equilibration estimate that does not use submeshes or a combination of estimators. The patch problems of Section 3 have no weights, but the evaluation (3.1) is a combination of two estimators, and its second entry carries the weight $\theta_K^{1/2}$ by Lemma 5.6. The weight is not needed in the construction because the unweighted minimiser is already small in the weighted norm $\varepsilon^{-1}\varpi_a^2\|\tau^a\|^2$; this is case 1 of Lemma 5.9. Without the minimum in (3.1) robustness is lost, in accordance with [SV20].

5.3 Local flux bounds

For a face F, let $j_F = [\![D\nabla u_h\cdot n]\!]$ if F is interior and $j_F = D\nabla u_h\cdot n - \Pi_F g_\ell$ if $F\subset\Gamma_N$, and let $J_a^2 = \sum_{F\in\mathcal{F}_a}\|j_F\|_F^2$.

Lemma 5.8 (conforming reconstruction)**.** *For each vertex a there is $v^\circ\in V^a$ with $\|\psi_a D\nabla u_h + v^\circ\|_{\omega_a} \leq C_B h_a^{1/2} J_a$.*

Proof. Define v° by its RTN_1 degrees of freedom: on each cell, the interior moments of $-\psi_a D\nabla u_h$; on interior faces of $\mathcal{T}_a$, the average of the two normal traces of $-\psi_a D\nabla u_h$; on $\partial\omega_a$, zero (where $\psi_a = 0$), the prescribed $-\Pi_F(\psi_a g_\ell)$ on Neumann faces, and the one-sided trace on free faces. Then $w = \psi_a D\nabla u_h + v^\circ$ is a broken RTN_1 field with vanishing interior moments. Its face moments are $\pm\frac{1}{2}\psi_a j_F$ on interior faces, $\psi_a j_F$ on Neumann faces and 0 otherwise; they are exact because $\psi_a D\nabla u_h\cdot n\in P_1(F)$. Scaling to the reference element gives $\|w\|_K \leq C h_K^{1/2}\sum_{F\subset\partial K}\|\psi_a j_F\|_F$, and $\|\psi_a\|_\infty = 1$. $\square$

Lemma 5.9 (weight-free patch bound)**.** *Let*

$$\mathrm{RES}_a^2 = \sum_{K\in\mathcal{T}_a}\alpha_K^2\|r_\ell\|_K^2 + \sum_{F\in\mathcal{F}_a}\varepsilon^{-1/2}\alpha_F\|j_F\|_F^2.$$

Then

$$\varepsilon^{-1}\varpi_a^2\|\tau^a\|_{\omega_a}^2 + \sigma^{-1}\|m^a\|_{\omega_a}^2 \leq C\big(\mathrm{RES}_a^2 + [\theta_a<1]\,\sigma^{-1}c_a^2|\omega_a|\big),$$

where $[\theta_a<1]$ is 1 if $\theta_a<1$ and 0 otherwise. For $\sigma = 0$ the second term on the left and the bracket are absent.

Proof in the diffusion regime ($\theta_a\geq 1$, or $\sigma = 0$). Here $\varpi_a = 1$. By Lemma 5.2 there is $v^*\in V^a$ with $\operatorname{div} v^* = \tilde{g}_a$; the shift makes the data compatible, and the constant c_a drops out against test functions with mean zero. Elementwise integration by parts, with $\operatorname{div}(\psi_a D\nabla u_h)|_K = \nabla\psi_a\cdot D\nabla u_h$ ($D\nabla u_h$ is constant on K), gives

$$\|\psi_a D\nabla u_h + v^*\| \leq C\sup_{\substack{w\in H^1_\dagger(\omega_a)\\ \|\nabla w\|=1}}\Big[(\Pi_1(\psi_a r_\ell), w) + \sum_F(\psi_a j_F, w)_F\Big].$$

The Poincaré inequality on ω_a (the Friedrichs inequality on patches with a free face) and the trace inequality give $\|\psi_a D\nabla u_h + v^*\| \le C(h_a\|r_\ell\|_{\omega_a} + h_a^{1/2}J_a)$. Hence

$$J_a(\sigma^a) \le J_a(v^*) = \varepsilon^{-1}\|\psi_a D\nabla u_h + v^*\|^2 \le C(\varepsilon^{-1}h_a^2\|r_\ell\|^2 + \varepsilon^{-1}h_a J_a^2).$$

Since $\theta_a \ge 1$, $\alpha_K \simeq h_a\varepsilon^{-1/2}$ and $\alpha_F \simeq h_a\varepsilon^{-1/2}$, so the right-hand side is bounded by $C\,\mathrm{RES}_a^2$, and the left-hand side of the lemma is bounded by $J_a(\sigma^a)$. For $\sigma = 0$, v^* is admissible for the constrained problem. □

Proof in the reaction regime ($\theta_a < 1$). The competitor of [SV20] does not suffice here: its flux part costs $\varepsilon^{-1}h_a J_a^2$, which is θ_a^{-1} times the efficient size $\varepsilon^{-1/2}\sigma^{-1/2}J_a^2$. In [SV20] this is compensated by the weight $\varpi_a^2 \simeq \theta_a$ in the construction. Here we compare with the best approximation instead.

Let $v_b = \operatorname{argmin}_{v\in V^a}\|\psi_a D\nabla u_h + v\|$, without divergence condition, and $\rho = \|\psi_a D\nabla u_h + v_b\|$. Since $\sigma^a \in V^a$, $\|\tau^a\| \ge \rho$. The optimality $J_a(\sigma^a) \le J_a(v_b)$ reads

$$\varepsilon^{-1}\|\tau^a\|^2 + \sigma^{-1}\|m^a\|^2 \le \varepsilon^{-1}\rho^2 + \sigma^{-1}\|m_b\|^2, \qquad m_b = \operatorname{div} v_b - \tilde{g}_a.$$

Equilibration term. Since $\rho \le \|\tau^a\|$, this gives $\sigma^{-1}\|m^a\|^2 \le \sigma^{-1}\|m_b\|^2$.

Bound on m_b. We have

$$m_b = \operatorname{div}_h(\psi_a D\nabla u_h + v_b) - \Pi_1(\psi_a r_\ell) + c_a.$$

Lemma 5.1 gives $\|\operatorname{div}_h(\psi_a D\nabla u_h + v_b)\| \le Ch_a^{-1}\rho$, and Lemma 5.8 gives $\rho \le \|\psi_a D\nabla u_h + v^\circ\| \le Ch_a^{1/2}J_a$. Hence

$$\sigma^{-1}\|m_b\|^2 \le C(\sigma^{-1}h_a^{-1}J_a^2 + \sigma^{-1}\|r_\ell\|_{\omega_a}^2 + \sigma^{-1}c_a^2|\omega_a|).$$

Since $\theta_a < 1$, $\alpha_F \simeq \sigma^{-1/2}$ and $\alpha_K \simeq \sigma^{-1/2}$, so $\sigma^{-1}h_a^{-1}J_a^2 = \theta_a\varepsilon^{-1/2}\sigma^{-1/2}J_a^2 \le C\varepsilon^{-1/2}\alpha_F J_a^2$ and $\sigma^{-1}\|r_\ell\|^2 \le C\sum_K \alpha_K^2\|r_\ell\|_K^2$.

Flux term. From the optimality, $\|\tau^a\|^2 \le \rho^2 + (\varepsilon/\sigma)\|m_b\|^2$. Multiplying by $\varepsilon^{-1}\varpi_a^2 = \varepsilon^{-1}\theta_a$,

$$\varepsilon^{-1}\theta_a\|\tau^a\|^2 \le \theta_a\varepsilon^{-1}\rho^2 + \theta_a\sigma^{-1}\|m_b\|^2.$$

The first term is bounded by $C\theta_a\varepsilon^{-1}h_a J_a^2 = C\varepsilon^{-1/2}\sigma^{-1/2}J_a^2$, and the second by θ_a times the bound on m_b. Both are bounded by $C(\mathrm{RES}_a^2 + \sigma^{-1}c_a^2|\omega_a|)$. □

5.4 Defect terms and residual bounds

Lemma 5.10 (defect terms)**.**

(i) For a compatible vertex a with $\theta_a < 1$, $\sigma^{-1}c_a^2|\omega_a| \le C\|\ell\|_{*,\omega_a}^2$.

(ii) The functional $\ell_{\mathrm{def}}(v) = (\psi_{\mathrm{def}}, v)$ satisfies $\|\ell_{\mathrm{def}}\|_* \le C_I\|\ell\|_*$. Under Assumption 2.1, also $\eta_{\ell,\mathrm{def}} \le C_\Pi C_I\|\ell\|_*$.

(iii) For $\ell = \mathrm{Res}_A$, $\ell(\psi_a) = \delta_a^{\mathrm{alg}} + \delta_a^{\mathrm{quad}}$. At the discrete solution, $\|\ell_{\mathrm{def}}\|_* = \mathrm{osc}_q$, and under Assumption 2.1 both defect terms are bounded by $C\,\mathrm{osc}_q$; they vanish when the data are integrated exactly. At an iterate, the local defect term is bounded by $C\|\,\mathrm{Res}_A\,\|_{*,\omega_a}$ by (i), and $\|\ell_{\mathrm{def}}\|_* \le C_I\|\,\mathrm{Res}_A\,\|_*$ by (ii); under Assumption 2.1 also $\eta_{\ell,\mathrm{def}} \le C_\Pi C_I\|\,\mathrm{Res}_A\,\|_*$.

Proof. (i) For compatible a, $\psi_a \in H^1_*(\omega_a)$, so $|c_a||\omega_a| = |\ell(\psi_a)| \le \|\ell\|_{*,\omega_a}|||\psi_a|||_{\omega_a}$, and $|||\psi_a|||_{\omega_a}^2 \le C(\varepsilon h_a^{-2} + \sigma)|\omega_a| = C\sigma(\theta_a^2+1)|\omega_a| \le 2C\sigma|\omega_a|$.

(ii) We have $\ell_{\mathrm{def}}(v) = \sum_{a\ \mathrm{compatible}} c_a(\chi_{\omega_a}, v) = \sum_a \ell(\psi_a)\bar{v}_{\omega_a} = \ell(I_c v)$, and Lemma 5.4 gives the first claim. For the second, let (s, z) attain the continuous dual norm: $\psi_{\mathrm{def}} = \operatorname{div} s + z$ with

$\|\ell_{\mathrm{def}}\|_*^2 = \varepsilon^{-1}\|s\|^2 + \sigma^{-1}\|z\|^2$; this is the duality for the symmetric operator $-\varepsilon\Delta + \sigma$, and $z = 0$ if $\sigma = 0$. Then $s_h = \Pi_h s$ is admissible for the discrete minimum, because $\operatorname{div} s_h = \Pi_0 \operatorname{div} s = \psi_{\mathrm{def}} - \Pi_0 z$ (recall $\psi_{\mathrm{def}} \in P_0$). Hence $\eta_{\ell,\mathrm{def}}^2 \le \varepsilon^{-1} C_\Pi^2 \|s\|^2 + \sigma^{-1}\|z\|^2 \le C_\Pi^2 \|\ell_{\mathrm{def}}\|_*^2$.

(iii) follows from Remark 3.1 and (ii). □

Lemma 5.11 (residual estimator and local dual norms [V05, proof of Thm. 4.1])**.** *For every cell K and every face F,*

$$\alpha_K \|\Pi_1 r_\ell\|_K \le C(\|\ell\|_{*,K} + \alpha_K \|r_\ell - \Pi_1 r_\ell\|_K),$$
$$\varepsilon^{-1/4} \alpha_F^{1/2} \|j_F\|_F \le C(\|\ell\|_{*,\omega_F} + \alpha_F \|r_\ell\|_{\omega_F} + \varepsilon^{-1/4} \alpha_F^{1/2} \|g_\ell - \Pi_F g_\ell\|_F),$$

where the last term is present only for $F \subset \Gamma_N$.

Proof. For $w \in P_1(K)$, $\ell(b_K w) = (r_\ell, b_K w)_K$, since $b_K = 0$ on ∂K and $\operatorname{div}(D\nabla u_h) = 0$ in K. Take $w = \Pi_1 r_\ell$ and apply Lemma 5.3. For the face term, $\ell(b_F^\theta \mu) = (r_\ell, b_F^\theta \mu)_{\omega_F} + (j_F, b_F^\theta \mu)_F$, plus the Neumann data oscillation on $F \subset \Gamma_N$, where $b_F^\theta \mu \in H_*^1(\omega_F)$ does not vanish on F. Take $\mu = j_F$, apply Lemma 5.3 and use the element bound. Only the form (2.1) of ℓ is used; neither the discrete equation nor the bilinear form B enters. □

6 Proofs of the main results

Proof of Theorem 4.1. For any $\sigma_h \in H(\operatorname{div})(\Omega)$ with $\sigma_h \cdot n = -\Pi_F g_\ell$ on Γ_N, and $\tau = D\nabla u_h + \sigma_h$, integration by parts gives

$$\ell(v) = (r_\ell - \Pi_1 r_\ell, v) + (\Pi_1 r_\ell - \operatorname{div} \sigma_h, v) - (\tau, \nabla v) + (g_\ell - \Pi_F g_\ell, v)_{\Gamma_N}.$$

By construction $\sum_a \tilde{g}_a = \Pi_1 r_\ell - \psi_{\mathrm{def}}$, so $\Pi_1 r_\ell - \operatorname{div} \sigma_h = \psi_{\mathrm{def}} - \sum_a m^a$. We bound each term.

- $(\tau, \nabla v)_K$: either by the Cauchy–Schwarz inequality, or by the identity $(\tau, \nabla v)_K = (\tau \cdot n, v - \bar{v}_K)_{\partial K} - (\operatorname{div} \tau, v - \bar{v}_K)_K$, together with the trace inequality $\|w\|_F^2 \le \frac{|F|}{|K|}(\|w\|^2 + \frac{2}{d} h_K \|w\| \|\nabla w\|)$ and the Poincaré inequality. This gives the minimum in (3.1).
- $(\sum_a m^a, v)_K \le \sigma^{-1/2} \|\sum_a m^a\|_K |||v|||_K$.
- The oscillation term by the Poincaré inequality, with the constant $\min\{h_K/(\pi\varepsilon^{1/2}), \sigma^{-1/2}\} = m_K$.
- (ψ_{def}, v): for any admissible s, write $\psi_{\mathrm{def}} = \operatorname{div} s + z$; then $(\psi_{\mathrm{def}}, v) = -(s, \nabla v) + (z, v) \le (\varepsilon^{-1}\|s\|^2 + \sigma^{-1}\|z\|^2)^{1/2} |||v|||$.
- The Neumann term by η_N (for $\ell = -\tilde{S}$, $g_\ell = 0$).

This proves $\|\operatorname{Res}_A\|_* \le \eta_A + \eta_N$ and $\|\tilde{S}\|_* \le \eta_{-\tilde{S}}$; the bound by η_S^{alt} holds by assumption. Then $\|\operatorname{Res}\|_* \le \|\operatorname{Res}_A\|_* + \|\tilde{S}\|_* \le \eta$, and Lemma 2.3 gives the rest. □

Proof of Theorem 4.2. Step 1. On K, $\tau = \sum_{a \in \mathcal{V}_K} \tau^a$, and $\varpi_K \simeq \varpi_a$ for $a \in \mathcal{V}_K$ by shape regularity. Lemma 5.6 and $m_K \le \alpha_K$ give

$$\eta_{\ell,K} \le C \sum_{a \in \mathcal{V}_K} [\varepsilon^{-1/2} \varpi_a \|\tau^a\|_K + \sigma^{-1/2} \|m^a\|_K] + \alpha_K \|r_\ell - \Pi_1 r_\ell\|_K.$$

Step 2. Lemma 5.9 for each a, and Lemma 5.10(i) for the local defect, give

$$\eta_{\ell,K}^2 \le C \sum_{a \in \mathcal{V}_K} (\mathrm{RES}_a^2 + \|\ell\|_{*,\omega_a}^2) + \operatorname{osc}_K(\ell)^2.$$

Step 3. Lemma 5.11, with $\|r_\ell\| \le \|\Pi_1 r_\ell\| + \|r_\ell - \Pi_1 r_\ell\|$, gives

$$\mathrm{RES}_a^2 \le C \sum_{K' \subset \omega_a,\ F \in \mathcal{F}_a} (\|\ell\|_{*,\omega_F}^2 + \|\ell\|_{*,K'}^2 + \mathrm{osc}_{K'}(\ell)^2).$$

This proves the local bound. *Step 4.* The sets K, ω_F and ω_a overlap at most $N(d,\vartheta)$ times, so Lemma 5.5 gives $\sum_K \eta_{\ell,K}^2 \le C(\|\ell\|_*^2 + \mathrm{osc}(\ell)^2)$. Lemma 5.10(ii) bounds $\eta_{\ell,\mathrm{def}}$ and gives the global bound. □

Proof of Corollary 4.3. The upper bound is Theorem 4.1. For the lower bound, $\eta_S \le \eta_{-\tilde{S}}$ and Theorem 4.2 give $\eta \le C(\|\operatorname{Res}_A\|_* + \|\tilde{S}\|_* + \mathrm{osc}) \le C(\|\operatorname{Res}\|_* + 2\|\tilde{S}\|_* + \mathrm{osc})$. By Lemma 2.3, $\|\operatorname{Res}\|_* \le \max\{1, c_b\}|||e|||_\oplus$, so $\eta_E \le (\kappa+1)\eta \le C\kappa \max\{1,c_b\}(E + \mathrm{osc}) \le C(1+c_b)^2(E+\mathrm{osc})$. □

Proof of Corollary 4.4. The upper bound follows from $E_c = (M-1)|||e||| + E \le (M-1)\eta + \eta_E$ (Theorem 4.1). For the lower bound, $\|\operatorname{Res}\|_* \le M|||e||| + \|B_A'(e,\cdot)\|_* = |||e|||_{\oplus,c}$, by the splitting in the proof of Lemma 2.3. With Theorem 4.2, $\eta \le C(|||e|||_{\oplus,c} + 2\|\tilde{S}\|_* + \mathrm{osc})$ and $\eta_E = \kappa\eta + \eta_S \le (\kappa+1)\eta$. Hence $\eta_{E,c} = (M-1)\eta + \eta_E \le (M+\kappa)\eta \le C(M+\kappa)(E_c + \mathrm{osc})$, and $M + \kappa = 2M + 2 \le 4(1+c_b)$. □

Remark 6.1 (where robustness comes from). The local efficiency constants depend only on d and ϑ; the global efficiency constant may also depend on C_Π in Assumption 2.1. The parameters ε, σ, b and c enter only through the weights α and θ, which cancel exactly in Lemmas 5.6, 5.9 and 5.10(i), and through the functional ℓ itself, which is never bounded in terms of the coefficients. The stabilisation tensor A appears only inside ℓ, through D. Hence the constants do not depend on the limiter.

Remark 6.2 (the role of Assumption 2.1). Assumption 2.1 is used only in Lemma 5.10(ii), the quasi-optimality of the global RT_0 defect bound, and through it for η_{alg} in Proposition 8.1. It is needed for $\sigma > 0$ and $\sigma = 0$ alike. The discrete inf-sup condition alone gives $\|s_h\|_{H(\mathrm{div})} \lesssim \|\psi_{\mathrm{def}}\|$, which is weaker than $\|s_h\| \lesssim \|\psi_{\mathrm{def}}\|_{H^{-1}}$. Without Assumption 2.1, Theorem 4.2 still holds for $\ell = \operatorname{Res}_A$ at the discrete solution, with $C_\Pi \operatorname{osc}_q$ replaced by the computed $\eta_{\mathrm{def},A}$; for $\ell = -\tilde{S}$, the true dual norm of the defect is still bounded by $C_I\|\tilde{S}\|_*$, but its RT_0 computation is not.

7 Local efficiency in a localisable error measure

Theorem 4.2 is local in the dual norms of the residual functionals. Converting these into the error brings in $\|B_A'(e,\cdot)\|_*$, a global dual norm that is not a sum of local contributions. We show that everything in E localises except one term, which depends only on u_h and is computed by the estimator. The tool is the localisation of dual norms of functionals that vanish on a partition of unity [BMV20].

7.1 Definitions

For a vertex a, let $\|\ell\|_{*,a} = \|\ell\|_{*,\omega_a}$, with the space $H_*^1(\omega_a)$ of Section 2. The *localised error measure* is

$$E_{\mathrm{loc}}^2 = |||e|||^2 + \sum_a (\|B_A'(e,\cdot)\|_{*,a}^2 + \|\tilde{S}\|_{*,a}^2),$$

and $E_{\mathrm{loc}}(\omega)$ denotes the same expression with the integrals and sums restricted to ω and to the patches in ω. The *nodal moment functional* of $\ell \in \{\operatorname{Res}_A, \tilde{S}\}$ is $\ell_{\mathrm{def}}(v) = \sum_{a \text{ compatible}} \ell(\psi_a)\,\bar{v}_{\omega_a}$. For $\tilde{S}$ it is built from the nodal moments $S(u_h; u_h, \psi_a)$ of the stabilisation. For Res_A it reduces to the quadrature defect at the discrete solution (Remark 3.1). By the proof of Theorem 4.1, $\|\ell_{\mathrm{def}}\|_* \le \eta_{\ell,\mathrm{def}}$; under Assumption 2.1 the converse holds up to C_Π (Lemma 5.10(ii)).

7.2 Results

Lemma 7.1 (robust localisation)**.** *For every $\ell \in H^1_D(\Omega)'$, $\sum_a \|\ell\|^2_{*,a} \le (d+1)\|\ell\|^2_*$. If, moreover, ℓ_{def} is defined from the nodal values $\ell(\psi_a)$ as above, then*

$$\|\ell\|_* \le C_L\Big(\sum_a \|\ell\|^2_{*,a}\Big)^{1/2} + \|\ell_{\mathrm{def}}\|_*,$$

with C_L depending only on d and ϑ, and not on ε or σ.

Proof. The first inequality is Lemma 5.5 with $N = d+1$, the maximal number of vertex patches covering a point. For the second, let $v \in H^1_D(\Omega)$, let $\bar v_a$ be the mean of v over ω_a for compatible a, and $\bar v_a = 0$ otherwise. Since $\sum_a \psi_a = 1$,

$$\ell(v) = \sum_a \ell(\psi_a(v-\bar v_a)) + \sum_{a \text{ compatible}} \bar v_a\, \ell(\psi_a) = \sum_a \ell(\psi_a(v-\bar v_a)) + \ell_{\mathrm{def}}(v),$$

and $\psi_a(v-\bar v_a) \in H^1_*(\omega_a)$. We show $|||\psi_a(v-\bar v_a)|||_{\omega_a} \le C|||v|||_{\omega_a}$. With $w = v - \bar v_a$, $\varepsilon\|\nabla(\psi_a w)\|^2 \le 2\varepsilon(\|\nabla w\|^2 + \|\nabla\psi_a\|^2_\infty \|w\|^2)$; the Poincaré inequality (compatible patches) or the Friedrichs inequality (patches with a free face) gives $\|w\| \le Ch_a\|\nabla v\|_{\omega_a}$, and $\|\nabla\psi_a\|_\infty \le Ch_a^{-1}$. For the reaction part, $\sigma\|\psi_a w\|^2 \le 2\sigma(\|v\|^2 + |\omega_a|\bar v_a^2) \le 4\sigma\|v\|^2_{\omega_a}$. Hence

$$\Big|\sum_a \ell(\psi_a(v-\bar v_a))\Big| \le C\Big(\sum_a \|\ell\|^2_{*,a}\Big)^{1/2}\Big(\sum_a |||v|||^2_{\omega_a}\Big)^{1/2} \le C\sqrt{d+1}\Big(\sum_a \|\ell\|^2_{*,a}\Big)^{1/2}|||v|||. \qquad \square$$

Theorem 7.2, part 1, applies the first inequality to $B'_A(e,\cdot)$, which has no partition-of-unity structure; parts 2 and 3 use the second inequality only for Res_A and $\tilde S$. On a patch, the error does not vanish on the boundary, so no Friedrichs inequality is available for it; the local conversion therefore needs the σ-based constant

$$c_{b,\sigma} := \|c - \tfrac12 \operatorname{div} b\|_\infty/\sigma \quad (\sigma > 0), \qquad c_{b,\sigma} := 0 \quad \text{if } c = \tfrac12 \operatorname{div} b.$$

The remaining case, $\sigma = 0$ with $c \ne \frac12 \operatorname{div} b$, is treated in Remark 7.3.

Theorem 7.2 (localisable two-sided bounds)**.** *Assume (H1)–(H6), and $\sigma > 0$ or $c = \frac12 \operatorname{div} b$.*

1. ***Guaranteed upper bound:*** *$E_{\mathrm{loc}} \le \sqrt{d+1}\,E \le \sqrt{d+1}\,\eta_E$.*
2. ***Robust local efficiency:*** *$\eta_{A,K} + \eta^{\mathrm{patch}}_{S,K} \le C(1+c_{b,\sigma})(E_{\mathrm{loc}}(\tilde\omega_K) + \mathrm{osc}(\tilde\omega_K))$, where $\eta^{\mathrm{patch}}_{S,K}$ is the cell indicator of $-\tilde S$, without the global defect term.*
3. ***Equivalence up to computable terms:*** *$E \le C(1+c_{b,\sigma})^2(E_{\mathrm{loc}} + \eta_{\mathrm{def},S} + \eta_{\mathrm{def},A})$. This bound does not need Assumption 2.1.*

At the discrete solution, $\|(\mathrm{Res}_A)_{\mathrm{def}}\|_* = \mathrm{osc}_q$, and under Assumption 2.1 also $\eta_{\mathrm{def},A} \le C_\Pi\,\mathrm{osc}_q$; both vanish for exactly integrated data. Hence $E_{\mathrm{loc}} \lesssim E \lesssim E_{\mathrm{loc}} + \eta_{\mathrm{def},S} + \eta_{\mathrm{def},A}$: in the global measure E, the only part that does not localise is the dual norm of the nodal moments of the stabilisation, a function of u_h alone that the estimator computes.

Proof. Part 1 is the first inequality of Lemma 7.1, applied to $B'_A(e,\cdot)$ and to $\tilde S$, together with Theorem 4.1.

Part 2. The local bounds of Theorem 4.2 use $\|\ell\|_{*,K}$ and $\|\ell\|_{*,\omega_F}$, which are bounded by $\|\ell\|_{*,a}$ for $a \in K$ or $a \in F$. For $\ell = \mathrm{Res}_A$, $\|\,\mathrm{Res}_A\,\|_{*,a} \le \|\,\mathrm{Res}\,\|_{*,a} + \|\tilde S\|_{*,a}$, and $\|\,\mathrm{Res}\,\|_{*,a} \le \max\{1, c_{b,\sigma}\}|||e|||_{\omega_a} + \|B'_A(e,\cdot)\|_{*,a}$, because $|B'_S(e,v)| \le \max\{1,c_{b,\sigma}\}|||e|||_{\omega_a}|||v|||_{\omega_a}$ for $v \in H^1_*(\omega_a)$;

this uses $\|e\|_{\omega_a} \le \sigma^{-1/2}|||e|||_{\omega_a}$. For $\ell = -\tilde{S}$, the local defect term of Lemma 5.10(i) is bounded by $C\|\tilde{S}\|_{*,a}$.

Part 3. By Lemma 2.3, $E \le \kappa\|\operatorname{Res}\|_* + \|\tilde{S}\|_* \le \kappa\|\operatorname{Res}_A\|_* + (\kappa+1)\|\tilde{S}\|_*$. Apply the second inequality of Lemma 7.1 to Res_A and to $\tilde{S}$, bound $\|\operatorname{Res}_A\|_{*,a}$ as in part 2, and use $\|\ell_{\mathrm{def}}\|_* \le \eta_{\ell,\mathrm{def}}$. With $c_b \le c_{b,\sigma}$, the constant is $C\kappa\max\{1, c_{b,\sigma}\} \le C(1+c_{b,\sigma})^2$. □

Remark 7.3 ($\sigma = 0$ with $c \ne \frac{1}{2}\operatorname{div} b$). For $v \in H^1_*(\omega_a)$, v vanishes on $\partial\omega_a \setminus \Gamma_N$, which has positive measure, so $\|v\|_{\omega_a} \le C_P h_a \|\nabla v\|_{\omega_a}$ with $C_P = C_P(d,\vartheta)$. Hence

$$\big|((c - \tfrac{1}{2}\operatorname{div} b)e, v)_{\omega_a}\big| \le \mu_a |||v|||_{\omega_a}, \qquad \mu_a := C_P \|c - \tfrac{1}{2}\operatorname{div} b\|_{\infty,\omega_a} h_a \varepsilon^{-1/2} \|e\|_{\omega_a},$$

and $\|\operatorname{Res}\|_{*,a} \le |||e|||_{\omega_a} + \mu_a + \|B'_A(e,\cdot)\|_{*,a}$. Part 2 of Theorem 7.2 therefore holds for $\sigma = 0$ with $c_{b,\sigma}$ replaced by 0 and E^2_{loc} augmented by $\sum_a \mu_a^2$. Part 3 holds with the same augmentation and the constant $C\kappa$, where $\kappa = 2 + \max\{1, c_b\}$ uses the Friedrichs-based c_b of (H3); this is admissible there because the first step of the proof of part 3 is global. Part 1 is not affected.

The additional term is a weighted L^2 norm of the error. By the Friedrichs inequality on $H^1_D(\Omega)$,

$$\Big(\sum_a \mu_a^2\Big)^{1/2} \le \sqrt{d+1}\, C_P C_F \, \|c - \tfrac{1}{2}\operatorname{div} b\|_\infty \, \frac{h_{\max}}{\varepsilon} \, |||e|||.$$

It is therefore dominated by $|||e|||$ when $h_{\max} \lesssim \varepsilon/(C_P C_F \|c - \frac{1}{2}\operatorname{div} b\|_\infty)$. In the convection-dominated regime $h \gg \varepsilon$ it is a genuinely stronger quantity, and the local efficiency statement is correspondingly weaker.

Remark 7.4 (how much weaker is E_{loc}?). By part 1, E_{loc} can be much smaller than E. If $\sigma = 0$ and $B'_A(e,\cdot)$ is smooth and of global extent, then $\|B'_A(e,\cdot)\|_* \sim \varepsilon^{-1/2}\|b\cdot\nabla e\|_{H^{-1}}$, while $\sum_a \|B'_A(e,\cdot)\|^2_{*,a} \sim \varepsilon^{-1} h^2 \|b\cdot\nabla e\|^2$. Part 3 shows that this can only happen when the nodal moments of the stabilisation, measured by $\eta_{\mathrm{def},S}$, are large, and this term is computed.

8 Adaptive stopping of the nonlinear iteration

Let u^i be any iterate of any solver for the algebraically stabilised system: a fixed-point iteration, Newton's method, or a truncated linear solve. We write $E(u^i)$ for the measure E at u^i, with $\tilde{S}$ and A those of the iterate.

The split. At an iterate, $\operatorname{Res}_A(\psi_a) = -r_a + \delta_a^{\mathrm{quad}}$, where r is the assembled algebraic residual (Remark 3.1). We define

$$\eta_{\mathrm{alg}} := \kappa\, \eta_{\mathrm{RT}}\Big(\sum_a \delta_a^{\mathrm{alg}} \chi_{\omega_a}/|\omega_a|\Big), \qquad \eta_{\mathrm{disc}} := \eta_E - \eta_{\mathrm{alg}},$$

where $\eta_{\mathrm{RT}}(\psi)$ denotes the global RT_0 bound of Section 3 applied to ψ in place of ψ_{def}. Thus η_{alg} is computed from the assembled residual alone, by one global RT_0 solve. It equals $\kappa\eta_{\mathrm{def},A}$ when the assembly integrates all data exactly. The stopping criterion is

$$\eta_{\mathrm{alg}} \le \gamma\, \eta_{\mathrm{disc}}. \tag{8.1}$$

Proposition 8.1 (adaptive stopping)**.** *Assume (H1)–(H6).*

1. ***Reliability at every iterate:*** $E(u^i) \le \eta_E(u^i) = \eta_{\mathrm{disc}} + \eta_{\mathrm{alg}}$.

2. ***Efficiency at every iterate.*** *Under Assumption 2.1, for every iterate of every solver,*

$$\eta_{\mathrm{disc}} + \eta_{\mathrm{alg}} \le C(1+c_b)^2\big(E(u^i) + \mathrm{osc}(u^i)\big), \qquad \eta_{\mathrm{alg}} \le \kappa C_\Pi\big(C_I \|\operatorname{Res}_A(u^i)\|_* + \mathrm{osc}_q\big),$$

and $\eta_{E,c}(u^i) \le C(1+c_b)(E_c(u^i) + \mathrm{osc}(u^i))$. *The constants are independent of* ε, σ, h *and the iteration index.*

3. ***The stopping criterion.*** *If (8.1) holds, then* $E(u^i) \le (1+\gamma)\,\eta_{\mathrm{disc}}$, *so the algebraic part is at most the fraction* $\gamma/(1+\gamma)$ *of the current bound.*

Proof. Part 1 is Theorem 4.1 at u^i; it uses only that u^i is a conforming function with the right boundary values. *Part 2.* Nothing in Theorem 4.2 uses the discrete equation; Lemma 5.11 uses only the form (2.1) of ℓ. At an iterate, Theorem 4.2 therefore gives $\eta_A(u^i) \le C(\|\operatorname{Res}_A(u^i)\|_* + \mathrm{osc})$, including the local shifts (Lemma 5.10(i)) and the global defect term (Lemma 5.10(ii)). The same argument for the spread of $\delta^{\mathrm{alg}} = \delta - \delta^{\mathrm{quad}}$ gives $\|\ell^{\mathrm{alg}}_{\mathrm{def}}\|_* \le C_I\|\operatorname{Res}_A(u^i)\|_* + \mathrm{osc}_q$, and with Assumption 2.1 the bound on η_{alg}. Further, $\operatorname{Res}(u^i)(v) = (f,v) + (g_N,v)_{\Gamma_N} - B(u^i,v) = B(u-u^i,v)$ for $v \in H^1_D(\Omega)$, so Lemma 2.3 applies to $u-u^i$. The rest is Corollary 4.3, and Corollary 4.4 for E_c, at u^i. *Part 3* follows from part 1. □

Remark 8.2 (no condition on γ). In the framework of Ern and Vohralík [EV13], the efficiency proof uses a smallness condition on γ to control the linearisation and algebraic estimators by the discretisation estimator. Here no such condition is needed, because Theorem 4.2 never uses the discrete equation: the iterate is a conforming function like any other, and $E(u^i)$ contains its algebraic error. Assumption 2.1 is needed only for $\eta_{\mathrm{alg}} \lesssim \|\operatorname{Res}_A(u^i)\|_*$; without it, the patch part of η_{disc} is still efficient, and η_{alg} is still a guaranteed bound of the dual norm of the spread residual.

Remark 8.3 (limitations).

- Part 3 is a statement about the stopped iterate. The stronger conclusion $E(u^i) \le C\,E(u_h)$, that the stopped iterate is as good as the discrete solution, would need $|||u_h - u^i||| \lesssim \eta_{\mathrm{alg}}$, a discrete stability estimate for the nonlinear operator. For algebraic stabilisations this is not available without monotonicity; existence of discrete solutions is known via Brouwer's fixed-point theorem [BJK16].
- $\eta_{\mathrm{disc}} \ge 0$ is not guaranteed in general, because $\eta_{\mathrm{def},A}$ bounds the spread of the full defect, including the quadrature part; part 3 does not need it.
- On patches with a free face, the algebraic residual of the vertex is absorbed by the patch flux rather than by η_{alg}. This concerns only boundary vertices next to Γ_D, and is harmless for all three parts.
- Quasi-optimal computational cost is proved in [HPSV21] for strongly monotone Lipschitz operators. The operators of algebraic stabilisations are in general not monotone, so this theory does not apply, and rate-optimal cost remains open.

9 Variable diffusion

We now let $\varepsilon \in L^\infty(\Omega)$ with $\varepsilon \ge \varepsilon_0 > 0$, and $|||v|||^2 = \|\varepsilon^{1/2}\nabla v\|^2 + \sigma\|v\|^2$. Let $\Pi_1\varepsilon$ be the cellwise L^2 projection of ε onto P_1, and $\varepsilon_{\mathrm{lo},K}$, $\varepsilon_{\mathrm{hi},K}$ the minimum and the maximum of ε on K.

Construction. The fluxes use $D = \Pi_1\varepsilon\, I + A$, so that $D\nabla u_h \in P_1(K)^d$. Since $\psi_a D\nabla u_h$ is quadratic, the patch problems use its RTN_1 interpolant $\Pi_{\mathrm{RT}}(\psi_a D\nabla u_h)$, defined by the normal moments against $P_1(F)$ and the interior moments against $P_0(K)^d$. Three properties of Π_{RT} are used:

- it reproduces RTN_1, so the data are unchanged when $D\nabla u_h$ is cellwise constant;
- $\sum_a \Pi_{\mathrm{RT}}(\psi_a D\nabla u_h) = \Pi_{\mathrm{RT}}(D\nabla u_h) = D\nabla u_h$, so that $\tau = D\nabla u_h + \sigma_h = \sum_a \tau^a$ with $\tau^a = \Pi_{\mathrm{RT}}(\psi_a D\nabla u_h) + \sigma^a$;
- $\operatorname{div}\Pi_{\mathrm{RT}} = \Pi_1 \operatorname{div}$ and $\Pi_{\mathrm{RT}}(\cdot)\cdot n|_F = \Pi_F((\cdot)\cdot n)$.

The remainder is the oscillation $\eta_{\varepsilon,K} = \|\varepsilon^{-1/2}(\varepsilon - \Pi_1\varepsilon)\nabla u_h\|_K$, which is added to the indicators of Res_A (and of the direct variant).

Lemma 9.1 (what P_1 test functions see)**.** *For $v \in P_1$,*

$$\big((\varepsilon - \Pi_1\varepsilon)\nabla u_h, \nabla v\big)_K = (\nabla u_h \cdot \nabla v)|_K \int_K (\varepsilon - \Pi_1\varepsilon) = 0.$$

Consequently:

(i) at the discrete solution, $\mathrm{Res}_A(\psi_a) = 0$ with $\Pi_1\varepsilon$ in D, up to the quadrature defect of Remark 3.1, whose diffusion part is $\sum_{K\ni a}(Q_K^{\mathrm{asm}}(\varepsilon) - Q_K^{\mathrm{est}}(\varepsilon))(\nabla u_h\cdot\nabla\psi_a)|_K$, with Q_K the values of $\int_K \varepsilon$ by the two rules;

(ii) for $v \in H^1_D(\Omega)$, $|\,\mathrm{Res}(v) - \mathrm{Res}_{\Pi_1\varepsilon}(v)| \le \sum_K \eta_{\varepsilon,K}|||v|||_K$;

(iii) for smooth ε, $\eta_{\varepsilon,K} \le Ch_K^2|\varepsilon|_{W^{2,\infty}(K)}\,\varepsilon_{\mathrm{lo},K}^{-1/2}\|\nabla u_h\|_K$.

Proof. The identity holds because ∇u_h and ∇v are constant on K and $\Pi_1\varepsilon$ has the cell mean of ε. Then (i) follows, (ii) is the Cauchy–Schwarz inequality, and (iii) is the approximation property of Π_1. □

By (iii), η_ε is one order above the error. With the cell mean $\bar\varepsilon_K$ in place of $\Pi_1\varepsilon$, the remainder would only be $O(h)\|\nabla \ln\varepsilon\|_\infty|||u_h|||_K$, of the order of the error itself.

Theorem 9.2 (variable diffusion)**.**

1. ***Reliability.*** *Theorem 4.1 and the upper bounds of Corollaries 4.3 and 4.4 hold, with $\eta_{\varepsilon,K}$ added to the indicators of Res_A, with ε replaced by $\varepsilon_{\mathrm{lo},K}$ in every Poincaré, trace and Friedrichs constant (m_K, c_F, the cut-off and the defect problem), and with $\|\varepsilon^{-1/2}\tau\|_K$ evaluated with ε pointwise. Nothing beyond $\varepsilon \in L^\infty$ and $\varepsilon \ge \varepsilon_0 > 0$ is assumed.*
2. ***Local efficiency.*** *Theorem 4.2 holds with the oscillation η_ε added, and with constants that also depend on the patch contrast $\kappa_\varepsilon(\tilde\omega_K) = \max_{\tilde\omega_K}\varepsilon / \min_{\tilde\omega_K}\varepsilon$. The element residual becomes $\hat r_\ell = r_\ell + \operatorname{div}_h(D\nabla u_h)$, with $\operatorname{div}_h(D\nabla u_h) = \nabla\Pi_1\varepsilon\cdot\nabla u_h \in P_0$.*

For ε piecewise constant on the mesh, $\varepsilon_{\mathrm{hi},K}/\varepsilon_{\mathrm{lo},K} = 1$, and only the contrast between the cells of a patch remains.

Proof. Part 1 follows from Lemma 9.1, from the Cauchy–Schwarz inequality weighted by ε pointwise, and from the inverse inequalities with $\varepsilon_{\mathrm{lo},K}$, which only enlarge the constants.

Part 2. Lemma 5.8 holds with the data $\Pi_{\mathrm{RT}}(\psi_a D\nabla u_h)$: the field $w = \Pi_{\mathrm{RT}}(\psi_a D\nabla u_h) + v^\circ$ is broken RTN_1, its face moments are $\pm\frac12\Pi_F(\psi_a j_F)$, and its interior moments vanish. Lemmas 5.2

and 5.9 need broken RTN_1 data, which this is. The integration by parts in the diffusion regime of Lemma 5.9 reads

$$(\tilde{g}_a, w) - (\Pi_{\mathrm{RT}}(\psi_a D\nabla u_h), \nabla w) = (\Pi_1(\psi_a \hat{r}_\ell), w) + \sum_F (\Pi_F(\psi_a j_F), w)_F$$

for w with mean zero, because $\operatorname{div}_h \Pi_{\mathrm{RT}}(\psi_a D\nabla u_h) = \nabla\psi_a \cdot D\nabla u_h + \Pi_1(\psi_a \operatorname{div}_h(D\nabla u_h))$. The reaction regime is unchanged. In Lemma 5.11, $\ell(b_K w) = (\hat{r}_\ell, b_K w)_K$, since now $\operatorname{div}(D\nabla u_h) \neq 0$, and the bubble argument is unchanged; since $\operatorname{div}_h(D\nabla u_h) \in P_0$, $\hat{r}_\ell - \Pi_1 \hat{r}_\ell = r_\ell - \Pi_1 r_\ell$, so the oscillation is unchanged. Finally, in Lemmas 5.6 and 5.9 the patch and cell weights and the norms of the competitors are compared using the extreme values of ε on $\tilde{\omega}_K$, and in the bubble estimates ε appears only through the local energy norms, at the price of the same ratio. □

Remark 9.3 (robustness in the contrast). The reliability hypotheses are minimal: $\varepsilon \in L^\infty$ with $\varepsilon \geq \varepsilon_0$ is required for the norm anyway. The efficiency constants proved here depend on the patch contrast. Results for other discretizations should not be conflated: modified residual estimators for nonconforming linear elements can have contrast-independent reliability without quasi-monotonicity [CHZ], and equilibrated estimators for discontinuous elements have been proved robust with respect to arbitrary coefficient jumps [CCZ]. For conforming approximations, Li [Li26] proves that no finite-range locally computable estimator can have both reliability and efficiency constants independent of the contrast M; their product grows at least like $M^{1/4}$ at a checkerboard cross-point. Since the bound of Theorem 9.2 is guaranteed, with reliability constant 1 for every contrast, this implies that its efficiency constant must grow at least like $M^{1/4}$ for checkerboard coefficients: the contrast dependence in part 2 cannot be removed in general. For quasi-monotone coefficients, robust bounds are known for residual estimators [BV00, Pe02]; whether the present estimator is robustly efficient in that class is open.

10 Numerical experiments

We test the estimator against the exact error. The questions are the ones raised by the theory:

(Q1) Is the bound guaranteed, $E \leq \eta_E$, in every run (Theorem 4.1)?

(Q2) Are the effectivity indices bounded independently of ε, σ and the stabilisation (Theorem 4.2, Corollary 4.3), and do they stay bounded under refinement?

(Q3) Does the dependence on c_b follow Corollaries 4.3 and 4.4?

(Q4) Is the estimator locally efficient in the localisable measure (Theorem 7.2)?

(Q5) Do the indicators drive adaptive refinement to the optimal rate?

(Q6) Are the bounds valid at inexact iterates, and does the stopping criterion of Proposition 8.1 save iterations without spoiling the error?

(Q7) Does the variable-diffusion version keep its effectivity, with $\eta_\varepsilon = O(h^2)$ (Theorem 9.2)?

10.1 Setup

Discretisation. Conforming P_1 elements on uniform meshes of the unit square ($n \times n$ squares, each cut into two triangles; $N = (n+1)^2$ vertices) or the unit cube (n^3 cubes, each cut into six tetrahedra), and on red–green refined meshes in the adaptive runs. The stabilisations are the AFC scheme with the limiter of [BJK17] (BJK), and the symmetrised monotone upwind-type algebraic stabilisation

of [Kn23] (SMUAS). The nonlinear problem is solved by the damped fixed-point iteration of [AJ27] up to $\|r\| \le 10^{-10}\sqrt{N}$, with at most $2 \cdot 10^4$ iterations, unless stated otherwise.

Test problems (all with b constant, so $\sigma = c$, $c_b = 1$ and $\kappa = 3$, except in Section 10.5).

- *smooth*: $u = \prod_i \sin(\pi x_i) + x_1^2 \prod_{i>1} x_i$, $b = (1, \frac{1}{2})$ or $(1, \frac{1}{2}, \frac{1}{4})$, Dirichlet data on $\partial\Omega$; f is computed from u. The solution has no layers, so its error is not dominated by layers, but for small ε the scheme is convection-dominated on every mesh used.
- *layers*: $u = \prod_i w(x_i)$ with $w(s) = s - (e^{(s-1)/\varepsilon} - e^{-1/\varepsilon})/(1 - e^{-1/\varepsilon})$, $b = (1, \dots, 1)$, homogeneous Dirichlet data; u has exponential layers of width ε at the outflow boundary $x_i = 1$.
- *variable diffusion*: $u = \phi(x_1) \prod_{i>1} 4x_i(1 - x_i)$, either with $\varepsilon(x) = \varepsilon\, k^{x_1}$, $\phi(s) = 4s(1 - s)$ (*smooth*), or with ε jumping from ε to $k\varepsilon$ at $x_1 = \frac{1}{2}$ and ϕ chosen such that the flux $\varepsilon\phi'$ is continuous (*jump*); k is the contrast.

Error measures. $|||e|||$ is computed with a quadrature of degree 8. The dual norms $\|B'_A(e, \cdot)\|_*$ and $\|\tilde{S}\|_*$, and the local dual norms of E_{loc}, are computed from their Riesz representatives in the P_2 space on the same mesh (on each vertex patch for E_{loc}). These are *lower* bounds of the true dual norms, so the reported effectivity indices are slight overestimates; they become crude when the dual norm concentrates in layers that the mesh does not resolve (see Section 10.4). We report the effectivity indices

$$I = \frac{\eta}{|||e|||}, \qquad I_\oplus = \frac{\kappa\eta}{|||e|||_\oplus}, \qquad I_S = \frac{\eta_S}{\|\tilde{S}\|_*}, \qquad I_E = \frac{\eta_E}{E},$$

of which I_E is the one covered by Corollary 4.3; $I_\oplus \ge 1$ and $I \ge 1$ are guaranteed too, but only $I_\oplus$ is expected to be robust.

10.2 Solutions and indicators

Figure 1 shows the discrete solution of the layers problem for $\varepsilon = 10^{-3}$ on the 32×32 mesh, the cell indicators $\eta_K = \eta_{A,K} + \eta_{S,K}$, and the local energy errors $|||e|||_K$. The mesh does not resolve the layers ($h \approx 4.4 \cdot 10^{-2} \gg \varepsilon$); the AFC solution is free of oscillations, and both the indicators and the error concentrate in the cells at the outflow boundary, with a decay of several orders of magnitude into the interior. The indicator is larger than $|||e|||_K$ in the layer cells: it measures the residual in the dual norm, which contains the convective part $\|B'_A(e, \cdot)\|_*$ of the error.

10.3 Uniform refinement

Table 1 and Figure 2 report uniform refinement from $n = 8$ to $n = 128$.

- **Smooth solution, $\varepsilon = 10^{-2}$.** E and η_E converge at the optimal rate 1 in h once the mesh resolves the solution; on the coarse meshes both decrease faster, because the stabilisation term $\|\tilde{S}\|_*$ dominates E there and decreases at a higher rate. The effectivity I_E lies between 3.15 and 3.29 for $\sigma = 0$ and $\sigma = 1$ alike. The energy-norm index I tends to 1.1: in the diffusion-dominated regime the energy norm and the dual norm of the residual coincide asymptotically.
- **Layers, $\varepsilon = 10^{-2}$.** The layers are resolved only from $n = 64$ on, and the convergence of E accelerates accordingly; I_E stays between 2.77 and 2.94.

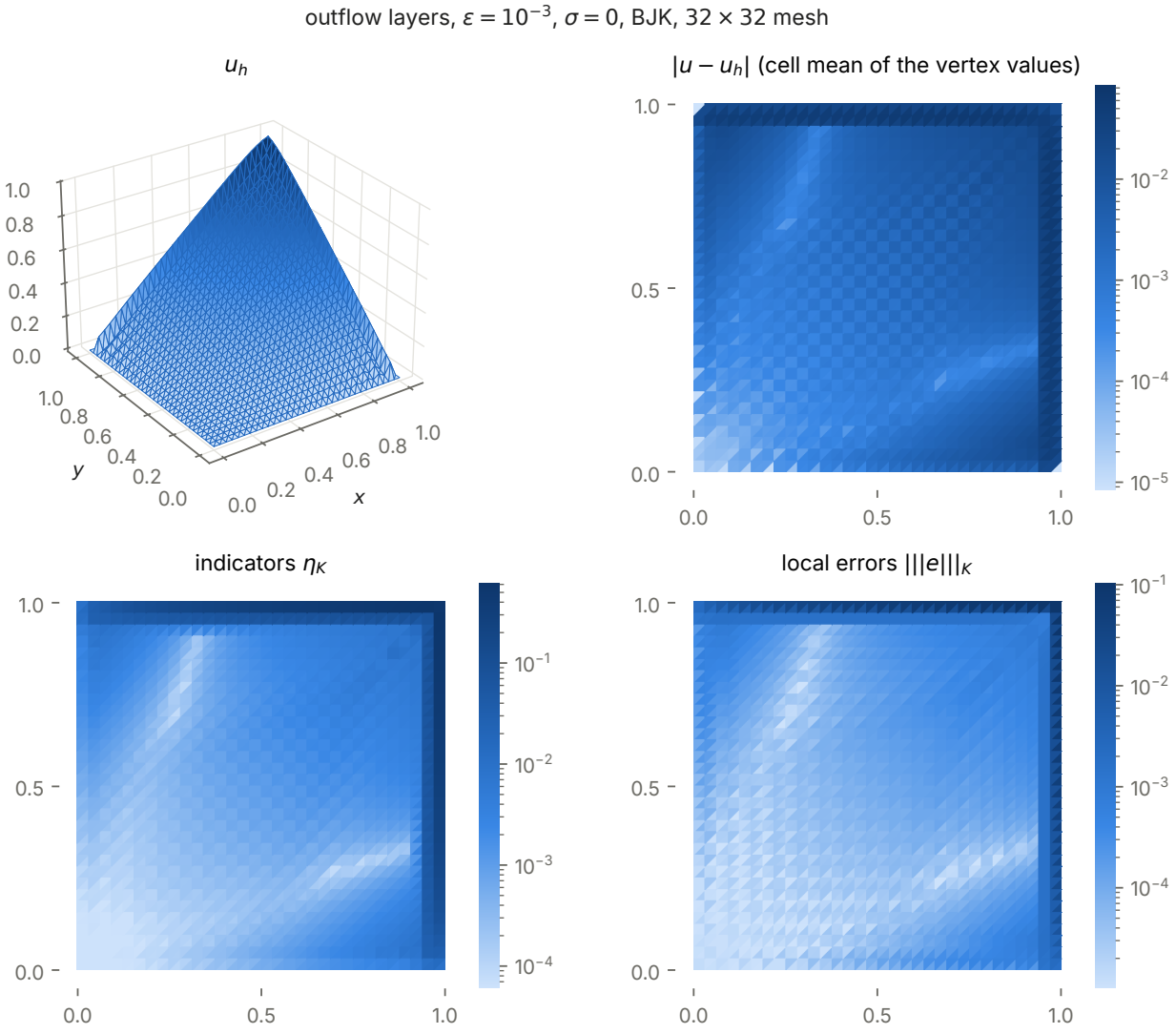


Figure 1: Layers problem, $\varepsilon = 10^{-3}$, $\sigma = 0$, BJK, $n = 32$: discrete solution u_h, indicators η_K and local errors $|||e|||_K$ (logarithmic colour scale, common to both).

- **Layers, $\varepsilon = 10^{-4}$.** The meshes never resolve the layers. The energy error $|||e|||$ *increases* under refinement, because the error, concentrated in the outflow cells, sees an increasing part of the layer, while E is dominated by the convective dual norm and decreases at the rate 0.45–0.53; η_E follows it with the rate 0.49–0.55. The index I_E is 3.07–3.31 on every mesh, whereas the energy-norm index I falls from 833 to 18.7: the energy-norm bound is guaranteed but useless in this regime, which is why the error is measured in E.

Table 1: Uniform refinement in 2D, BJK. Rates with respect to $N^{-1/2} \sim h$; the expected rate is 1 once the mesh resolves the solution. Effectivity indices $I = \eta/|||e|||$, $I_\oplus = \kappa\eta/|||e|||_\oplus$, $I_E = \eta_E/E$; the last column is the number of fixed-point iterations.

N	$\|\|\|e\|\|\|$	E	rate	η_E	rate	I	$I_\oplus$	I_E	it.
smooth, $\varepsilon = 10^{-2}$, $\sigma = 0$									
81	$4.88 \cdot 10^{-2}$	0.209		0.687		4.12	4.22	3.29	59
289	$2.15 \cdot 10^{-2}$	$4.48 \cdot 10^{-2}$	2.42	0.145	2.45	2.05	3.86	3.24	39
1089	$1.04 \cdot 10^{-2}$	$1.26 \cdot 10^{-2}$	1.91	$4.15 \cdot 10^{-2}$	1.89	1.31	3.46	3.28	18
4225	$5.16 \cdot 10^{-3}$	$5.51 \cdot 10^{-3}$	1.23	$1.76 \cdot 10^{-2}$	1.26	1.14	3.24	3.20	9
16641	$2.58 \cdot 10^{-3}$	$2.65 \cdot 10^{-3}$	1.07	$8.40 \cdot 10^{-3}$	1.08	1.09	3.18	3.17	6
smooth, $\varepsilon = 10^{-2}$, $\sigma = 1$									
81	$5.01 \cdot 10^{-2}$	0.178		0.582		3.34	4.04	3.27	56
289	$2.16 \cdot 10^{-2}$	$4.31 \cdot 10^{-2}$	2.23	0.136	2.29	1.89	3.69	3.15	37
1089	$1.04 \cdot 10^{-2}$	$1.25 \cdot 10^{-2}$	1.87	$4.12 \cdot 10^{-2}$	1.80	1.29	3.46	3.29	18
4225	$5.16 \cdot 10^{-3}$	$5.47 \cdot 10^{-3}$	1.22	$1.77 \cdot 10^{-2}$	1.24	1.15	3.29	3.24	9
16641	$2.58 \cdot 10^{-3}$	$2.64 \cdot 10^{-3}$	1.06	$8.45 \cdot 10^{-3}$	1.08	1.10	3.22	3.20	6
layers, $\varepsilon = 10^{-2}$, $\sigma = 0$									
81	0.522	2.6		7.62		4.19	3.78	2.94	67
289	0.471	1.64	0.72	4.72	0.75	2.88	3.64	2.88	47
1089	0.366	1	0.74	2.83	0.77	2.21	3.56	2.82	63
4225	0.231	0.401	1.35	1.15	1.33	1.52	3.26	2.86	28
16641	0.127	0.148	1.46	0.409	1.51	1.07	2.77	2.77	9
layers, $\varepsilon = 10^{-4}$, $\sigma = 0$									
81	$3.07 \cdot 10^{-2}$	$2.72 \cdot 10^{1}$		$9.02 \cdot 10^{1}$		833.01	4.76	3.31	405
289	$6.15 \cdot 10^{-2}$	$1.94 \cdot 10^{1}$	0.53	$6.37 \cdot 10^{1}$	0.55	290.45	4.81	3.28	709
1089	0.154	$1.42 \cdot 10^{1}$	0.47	$4.56 \cdot 10^{1}$	0.51	82.17	4.75	3.20	700
4225	0.242	$1.05 \cdot 10^{1}$	0.45	$3.27 \cdot 10^{1}$	0.49	37.26	4.65	3.12	626
16641	0.342	7.6	0.47	$2.33 \cdot 10^{1}$	0.49	18.72	4.54	3.07	566

10.4 Robustness in ε, σ and the stabilisation

Table 2 and Figure 4 sweep ε from 10^{-1} to 10^{-8} on the fixed mesh $n = 32$, for $\sigma \in \{0, 1\}$, both stabilisations and both problems.

- $\sigma = 0$**.** I_E lies in 2.80–3.23 for the layers problem and in 3.15–3.78 for the smooth problem, and becomes constant for $\varepsilon \le 10^{-5}$. $I_\oplus$ stays below 4.9, and the stabilisation part is estimated sharply, $I_S = 1.21$–1.34. BJK and SMUAS give the same indices I_E and $I_\oplus$ to within 3 %, as Remark 6.1 predicts: the limiter enters only through the functional, not the constants.
- **Energy norm.** The energy-norm index I grows like ε^{-1} for $\sigma = 0$, up to $2.5 \cdot 10^6$ at $\varepsilon = 10^{-8}$; the guaranteed energy-norm bound of [ABR17] behaves in the same way. For $\sigma = 1$, I saturates at about 60–90, the reaction giving L^2 control, while the ABR17 bound keeps growing ($1.3 \cdot 10^4$ at $\varepsilon = 10^{-8}$), because its reaction cut-off is not robust.

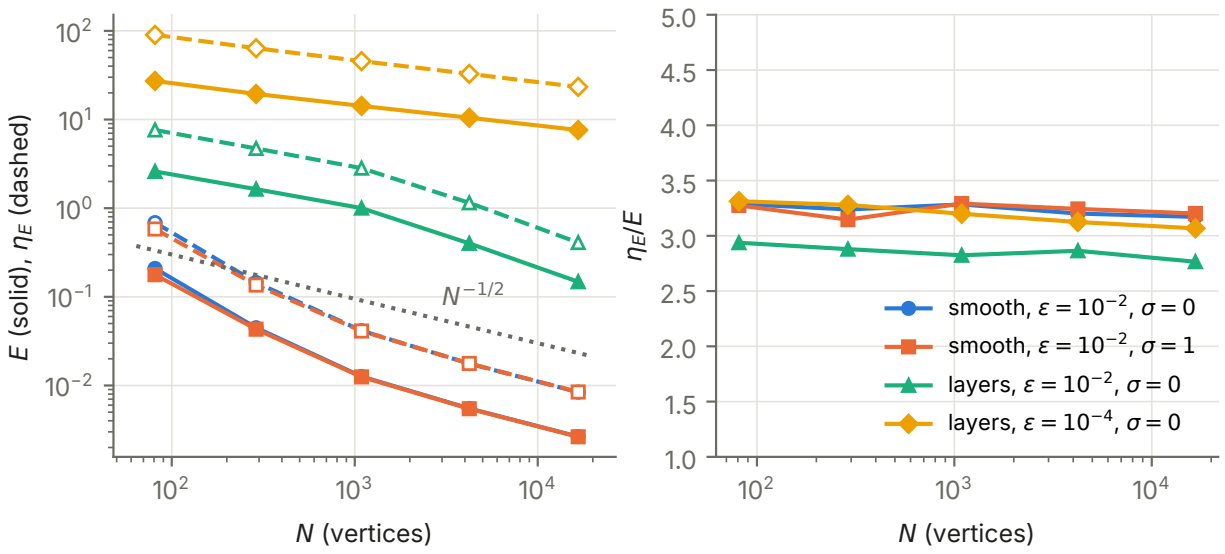


Figure 2: Uniform refinement in 2D, BJK. Left: E (solid) and η_E (dashed, open markers). Right: effectivity index $I_E = \eta_E/E$.

- $\sigma = 1$, **small** ε. Down to $\varepsilon = 10^{-6}$, I_E stays in 2.7–3.2 for the layers problem and in 3.2–4.9 for the smooth one. For $\varepsilon < 10^{-6}$ it rises, to 5.5 (layers) and 6.0 (smooth) at $\varepsilon = 10^{-8}$. This increase comes from I_S (7.8 and 10.4 at $\varepsilon = 10^{-8}$), while $I_\oplus$ *decreases*. This is consistent with a reference effect rather than with a loss of robustness: for $\sigma > 0$, the Riesz representative of $\tilde{S}$ has layers of width $(\varepsilon/\sigma)^{1/2} = 10^{-4}$, far below the mesh size $3 \cdot 10^{-2}$, so the same-mesh P_2 reference underestimates $\|\tilde{S}\|_*$. Theorem 4.2 bounds η_S by the true dual norm, independently of ε. The adaptive runs of Section 10.8 confirm this: once the mesh resolves the layers, I_S falls to 1.5–1.6 and I_E to about 3. The asymptotic reference below makes this quantitative.

An asymptotically exact reference for $\|\tilde{S}\|_*$. For $\theta \ll 1$ the dual norm of the stabilisation functional can be evaluated without resolving its Riesz representative. With $q = A\nabla u_h$ cellwise constant, $\tilde{S}(v) = (q, \nabla v) = -\sum_F (\llbracket q \cdot n \rrbracket, v)_F$ for $v \in H^1_D$, since $\operatorname{div} q = 0$ in each cell. Away from the vertices, the Riesz representative is a sum of one-dimensional layers $-\varepsilon z'' + \sigma z = \llbracket q \cdot n \rrbracket \, \delta_F$ across the faces, each of energy $\llbracket q \cdot n \rrbracket^2/(2(\varepsilon\sigma)^{1/2})$ per unit length. Hence

$$\|\tilde{S}\|_*^2 = \sum_{F \text{ interior}} \frac{\|\llbracket q \cdot n \rrbracket\|_F^2}{2(\varepsilon\sigma)^{1/2}} \, (1 + O(\theta)), \qquad \theta \to 0, \tag{10.1}$$

and the same sum over the faces interior to $\tilde{\omega}_K$ gives $\|\tilde{S}\|^2_{*,\tilde{\omega}_K}$. Table 3 and Figure 3 compare this value with P_2 references on the same mesh and on k-times refined meshes ($k = 2, 4, 8$), for the layers problem on the 16×16 mesh with $\sigma \in \{1, 10^4\}$, using the A+S estimator alone (without the direct variant of Remark 3.2).

- **The references.** At $\theta = 0.11$ the asymptotic value agrees with the refined reference to 1 % (2.92 against 2.95). For smaller θ the P_2 references are far from converged: at $\theta = 10^{-5}$ the same-mesh value underestimates $\|\tilde{S}\|_*$ by a factor 85, and even the 8 times refined one by a factor 30.

Table 2: Robustness in ε and σ on the uniform mesh with $n = 32$ ($N = 1089$). $I_E = \eta_E/E$, $I_\oplus = \kappa\eta/|||e|||_\oplus$, $I_S = \eta_S/\|\tilde{S}\|_*$, $I = \eta/|||e|||$ and, for comparison, the effectivity I_{ABR17} of the guaranteed energy-norm bound of [ABR17].

case	index	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}	10^{-8}
layers, BJK, $\sigma = 0$	I_E	2.96	2.82	3.06	3.20	3.23	3.23	3.23	3.23
	$I_\oplus$	2.98	3.56	4.30	4.75	4.84	4.84	4.84	4.84
	I_S	1.26	1.24	1.24	1.22	1.22	1.22	1.22	1.22
	I	1.06	2.21	7.95	82.17	$2.5 \cdot 10^3$	$2.5 \cdot 10^4$	$2.5 \cdot 10^5$	$2.5 \cdot 10^6$
	I_{ABR17}	1.47	1.93	6.26	64.53	$1.9 \cdot 10^3$	$1.9 \cdot 10^4$	$1.9 \cdot 10^5$	$1.9 \cdot 10^6$
layers, BJK, $\sigma = 1$	I_E	3.03	3.13	3.16	2.97	2.81	3.04	3.83	5.47
	$I_\oplus$	3.05	3.95	4.25	4.02	3.83	3.50	3.28	3.24
	I_S	1.29	1.29	1.41	1.54	1.66	2.55	4.41	7.79
	I	1.09	2.33	6.39	30.27	74.58	71.78	67.57	66.77
	I_{ABR17}	1.47	1.94	6.25	57.80	391.31	$1.3 \cdot 10^3$	$4.0 \cdot 10^3$	$1.3 \cdot 10^4$
layers, SMUAS, $\sigma = 0$	I_E	2.95	2.80	3.01	3.15	3.17	3.18	3.18	3.18
	$I_\oplus$	2.97	3.57	4.29	4.73	4.82	4.82	4.82	4.82
	I_S	1.34	1.24	1.25	1.23	1.23	1.23	1.23	1.23
	I	1.06	2.26	8.09	83.92	$2.7 \cdot 10^3$	$2.7 \cdot 10^4$	$2.7 \cdot 10^5$	$2.7 \cdot 10^6$
	I_{ABR17}	1.47	1.97	6.38	65.99	$2.2 \cdot 10^3$	$2.2 \cdot 10^4$	$2.2 \cdot 10^5$	$2.2 \cdot 10^6$
layers, SMUAS, $\sigma = 1$	I_E	3.03	3.10	3.10	2.91	2.73	2.97	3.78	5.42
	$I_\oplus$	3.04	3.95	4.20	4.00	3.81	3.49	3.27	3.24
	I_S	1.37	1.28	1.41	1.50	1.61	2.46	4.26	7.53
	I	1.08	2.36	6.14	28.08	66.99	64.21	60.54	59.85
	I_{ABR17}	1.47	1.99	6.38	58.84	389.93	$1.3 \cdot 10^3$	$4.0 \cdot 10^3$	$1.3 \cdot 10^4$
smooth, BJK, $\sigma = 0$	I_E	3.15	3.28	3.76	3.78	3.58	3.58	3.58	3.58
	$I_\oplus$	3.17	3.46	4.85	4.49	3.87	3.86	3.85	3.85
	I_S	1.29	1.30	1.24	1.21	1.22	1.22	1.22	1.22
	I	1.12	1.31	6.28	77.92	778.23	$6.9 \cdot 10^3$	$6.8 \cdot 10^4$	$6.8 \cdot 10^5$
	I_{ABR17}	1.70	1.76	5.18	74.26	842.71	$7.5 \cdot 10^3$	$7.4 \cdot 10^4$	$7.4 \cdot 10^5$
smooth, BJK, $\sigma = 1$	I_E	3.22	3.29	3.86	4.66	4.93	4.81	5.04	6.00
	$I_\oplus$	3.24	3.46	4.82	5.50	5.68	5.22	4.79	4.69
	I_S	1.32	1.37	1.45	1.66	2.24	3.41	5.88	10.39
	I	1.14	1.29	5.60	42.08	90.00	92.63	85.82	84.22
	I_{ABR17}	1.70	1.76	5.05	49.08	229.66	763.28	$2.4 \cdot 10^3$	$7.7 \cdot 10^3$

- **Theorem 4.2 for $\ell = -\tilde{S}$.** Measured against (10.1), the index $I_S = \eta_{-\tilde{S}}/\|\tilde{S}\|_*$ lies in 1.64–2.22 for all $\theta \le 10^{-2}$; the local ratio $\max_K \eta_{S,K}^2/\|\tilde{S}\|_{*,\tilde{\omega}_K}^2$ lies in 0.58–1.56. For this functional the theorem has no oscillation term, so this is a direct confirmation of robust global and local efficiency in the reaction regime.
- **The error measure.** With $\|\tilde{S}\|_*$ from (10.1) and the other two parts of E from the 8 times refined mesh, η_E/E stays in 8.3–10.2 over three decades of θ, whereas the same-mesh reference gives values up to 284. The rises of I_E in Table 2 and in Section 10.8 at small θ are therefore an artifact of the reference, not a loss of robustness.

Table 3: Reaction regime, layers problem, $n = 16$, BJK, the A+S estimator alone. $\|\tilde{S}\|_*$ from P_2 on the same mesh ($k = 1$), on the 8 times refined mesh ($k = 8$), and the asymptotic value (10.1); $I_S = \eta_{-\tilde{S}}/\|\tilde{S}\|_*$; η_E/E with the same-mesh reference and with E^*, in which $\|\tilde{S}\|_*$ is the larger of the $k = 8$ and the asymptotic value.

σ	ε	θ	$\|\tilde{S}\|_*$			I_S		η_E/E	
			$k=1$	$k=8$	asympt.	$k=8$	asympt.	$k=1$	E^*
1	10^{-2}	1.1	0.461	0.461	0.564	1.34	1.10	3.4	3.1
1	10^{-4}	$1.1 \cdot 10^{-1}$	2.67	2.95	2.92	1.72	1.73	5.6	5.1
1	10^{-6}	$1.1 \cdot 10^{-2}$	3.34	8.14	9.28	2.49	2.19	15.6	8.4
1	10^{-8}	$1.1 \cdot 10^{-3}$	3.35	9.77	29.4	5.33	1.77	43.5	9.3
10000	10^{-2}	$1.1 \cdot 10^{-2}$	5.02	11.9	13.5	2.52	2.22	14.0	8.3
10000	10^{-4}	$1.1 \cdot 10^{-3}$	5.29	15.2	45	5.36	1.81	30.4	9.0
10000	10^{-6}	$1.1 \cdot 10^{-4}$	5.29	15.2	142	15.68	1.68	91.1	9.9
10000	10^{-8}	$1.1 \cdot 10^{-5}$	5.29	15.2	450	48.39	1.64	283.6	10.2

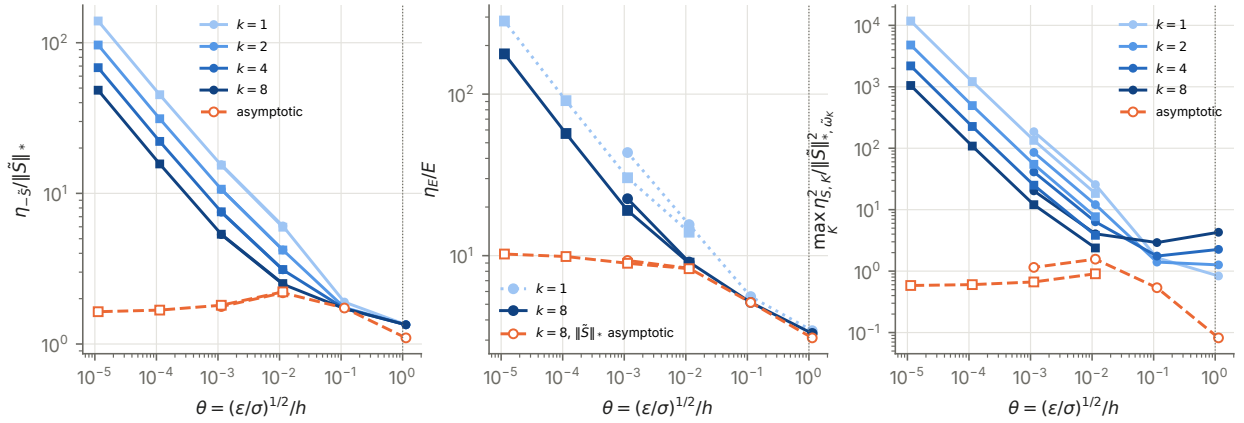


Figure 3: Reaction regime, layers problem, $n = 16$, A+S estimator; circles $\sigma = 1$, squares $\sigma = 10^4$. Left: $\eta_{-\tilde{S}}/\|\tilde{S}\|_*$ with P_2 references on k-times refined meshes and with the asymptotic value (10.1). Middle: η_E/E with the same-mesh and the 8 times refined references, and with $\|\tilde{S}\|_*$ from (10.1). Right: the local ratio for $\ell = -\tilde{S}$; the refined references are lower bounds of the local dual norms, so these ratios are upper estimates; the asymptotic value is valid only for $\theta \ll 1$.

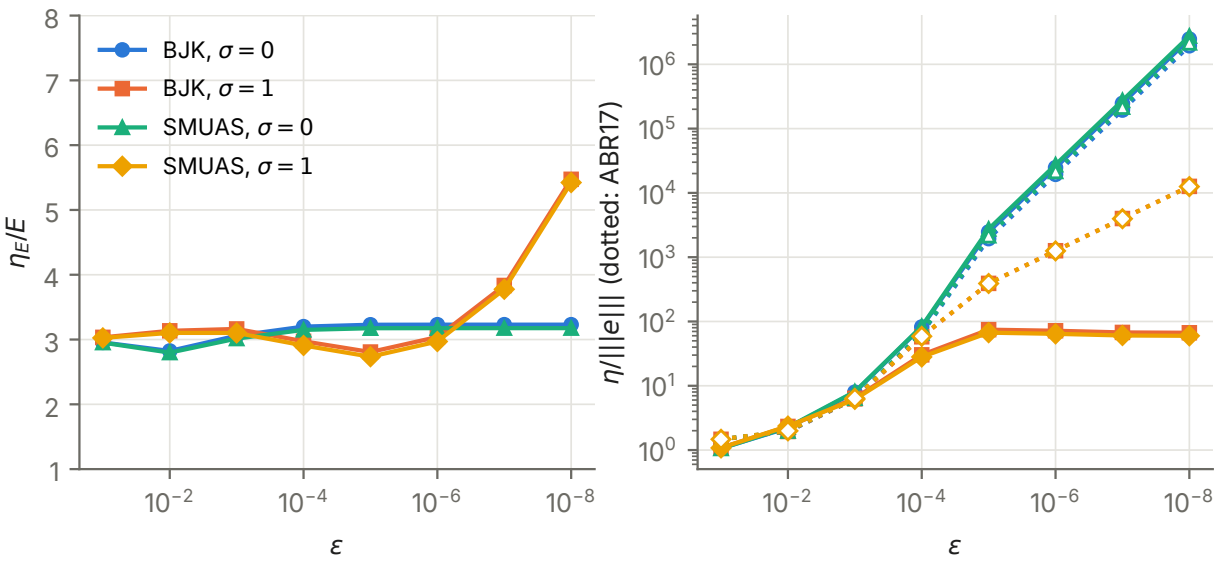


Figure 4: Layers problem, $n = 32$, as functions of ε. Left: $I_E = \eta_E/E$. Right: the energy-norm indices $\eta/|||e|||$ (solid) and, dotted with open markers, that of the guaranteed bound of [ABR17].

10.5 Variable reaction

With $c = 1 + (c_{\max} - 1)xy$ and $\sigma = 1$, the constant of (H3) is $\|c\|_\infty/\sigma = c_{\max}$, attained at the vertex $(1, 1)$. The computed c_b is the sampled enclosure of Remark 2.4, which is diagnostic: c is not affine on the cells, so the sampled maximum is enlarged by $\frac{h_K}{2}\max|\nabla c|$; here this gives $c_b = 1.12\,c_{\max}$, $1.06\,c_{\max}$ and $1.03\,c_{\max}$ for $n = 8, 16, 32$, and $\kappa = 2 + c_b$ (Table 4). The bound $E \le \eta_E$ holds in all runs. The index of $\eta_{E,c}$ grows at most linearly, $\eta_{E,c}/E_c \le 3\,c_b$ in all runs, as Corollary 4.4 predicts; it is often much smaller, down to $0.05\,c_b$, because the factor $M = c_b$ in front of $|||e|||$ in E_c also enlarges the error measure. The index of η_E grows faster, up to $5.2\,c_b$ at $c_b \approx 10^3$, but stays far below the worst case $(1 + c_b)^2$ of Corollary 4.3. A variable reaction with a small lower bound σ is therefore the one situation in which the estimator loses its robustness, in accordance with the theory; Section 11 discusses how to remove it.

Table 4: Variable reaction $c = 1 + (c_{\max} - 1)xy$, $\sigma = 1$, $\varepsilon \in \{10^{-3}, 10^{-6}\}$, $n \in \{8, 16, 32\}$: ranges over the six runs; c_b and κ are the means of the sampled enclosures (Remark 2.4; diagnostic), which exceed $c_{\max}$ by 3–12 %.

$c_{\max}$	c_b	κ	η_E/E	$\eta_{E,c}/E_c$	$\eta_{E,c}/(c_b E_c)$	$E \le \eta_E$
1	1.00	3.00	3.85–5.34	3.85–5.34	3.85–5.34	yes
10	10.66	12.66	14.63–19.94	8.69–30.35	0.85–2.95	yes
100	107.22	109.22	128–271	12.27–216	0.12–2.09	yes
1000	$1.1 \cdot 10^3$	$1.1 \cdot 10^3$	1618–5312	56.23–1125	0.05–1.09	yes

10.6 Local efficiency

Table 5 reports the localised measure for the layers problem on the 16×16 mesh. The local ratio $\max_K \eta_K^2/E_{\mathrm{loc}}(\tilde{\omega}_K)^2$ is at most 0.51 in all cases, and independent of ε for $\varepsilon \le 10^{-4}$: the

estimator is locally efficient, as stated in Theorem 7.2. The ratio E_{loc}/E lies between 0.48 and 1.01, so on these problems the localised measure captures most of the global one, and the guaranteed bound $E_{\mathrm{loc}} \le \sqrt{d+1}\,\eta_E$ of Theorem 7.2 overestimates E_{loc} by a factor 5–12. The largest value, $\eta_E/E_{\mathrm{loc}} = 7.0$ for $\sigma = 1$ and $\varepsilon = 10^{-8}$, is the same reference effect as in Section 10.4.

Table 5: Localised measure (Theorem 7.2), outflow layers, BJK, $n = 16$. The last but one column is $\max_K \eta_K^2/E_{\mathrm{loc}}(\tilde{\omega}_K)^2$ with $\eta_K = \eta_{A,K} + \eta_{S,K}$.

σ	ε	E_{loc}/E	η_E/E_{loc}	$\sqrt{d+1}\,\eta_E/E_{\mathrm{loc}}$	local max	I_E
0	10^{-2}	0.48	5.94	10.29	0.26	2.88
0	10^{-4}	0.60	5.50	9.53	0.37	3.28
0	10^{-6}	0.60	5.50	9.52	0.37	3.29
0	10^{-8}	0.60	5.50	9.52	0.37	3.29
1	10^{-2}	0.52	6.38	11.05	0.51	3.33
1	10^{-4}	0.95	3.07	5.32	0.13	2.93
1	10^{-6}	1.01	3.47	6.01	0.07	3.51
1	10^{-8}	1.01	7.04	12.20	0.07	7.13

10.7 Adaptive refinement

Starting from the 8×8 mesh, we mark by the indicators $\eta_K = \eta_{A,K} + \eta_{S,K}$ with the dual strategy of [AJ27] (all cells with $\eta_K \ge \theta \max \eta$, $\theta = 0.5$, reduced until at least 5 % of the cells are marked) and refine by red–green refinement, until $N > 3 \cdot 10^4$. Table 6 and Figures 5 and 6 show the results for the layers problem.

- **Effectivity.** I_E stays between 2.86 and 3.55 in every loop of all three runs, from $N = 81$ to $N = 45\,094$, while the local mesh size spans three orders of magnitude.
- **Where the mesh is refined.** The refinement concentrates in the outflow layers at $x = 1$ and $y = 1$ (Figure 6); the smallest cells have $h_{\min} = 3.5 \cdot 10^{-4}$ for $\varepsilon = 10^{-3}$, below ε, so the layers are resolved, and $h_{\min} = 1.7 \cdot 10^{-4}$ for $\varepsilon = 10^{-5}$, where they are not.
- **Convergence.** For $\varepsilon = 10^{-3}$, E decreases at the rate 1.8–1.9 with respect to $N^{-1/2}$ over the second half of the loops: this is the pre-asymptotic phase in which the layers become resolved, faster than the optimal rate 1. For $\varepsilon = 10^{-5}$ the rate is 0.98. On the finest uniform mesh ($N = 16\,641$) the error E is 18, 4.4 and 17 times larger than the adaptive error at the same N. Under uniform refinement E decreases at the rate 0.46–0.60 for $\sigma = 0$ and 0.04–0.51 for $\sigma = 1$, because the layers are never resolved; the index I_E is 2.86–3.37 there as well.
- **Energy norm.** Along the adaptive runs the energy-norm index I falls from 29 to 1.1 ($\varepsilon = 10^{-3}$) and from $8.4 \cdot 10^3$ to 6.2 ($\varepsilon = 10^{-5}$): once the mesh resolves the layers, the energy error and the dual norm of the residual become comparable, while I_E is insensitive to this.

10.8 Uniform versus adaptive refinement

We repeat the uniform-mesh studies of Tables 1 and 2 with adaptive refinement: the same problems, the same marking as in Section 10.7, from the 8×8 mesh up to $N \approx 2 \cdot 10^4$ (Table 1) and $N \approx 10^4$ (Table 2). For the robustness sweep we compare at equal cost: the adaptive loop whose N is closest to $N = 1089$, the size of the 32×32 mesh of Table 2.

Table 6: Adaptive refinement for the outflow-layers problem, BJK, dual marking by the SV indicators η_K, red–green refinement. The rate is that of E with respect to $N^{-1/2}$ over the second half of the loops; the last column is the ratio of E on the finest uniform mesh ($N = 16641$) to the adaptive E at the same N.

ε	σ	loops	N (last)	$h_{\min}$	E (last)	rate	η_E/E	gain
10^{-3}	0	15	33010	$3.45 \cdot 10^{-4}$	$7.395 \cdot 10^{-2}$	1.88	2.86–3.25	18.3
10^{-5}	0	12	31866	$1.73 \cdot 10^{-4}$	3.941	0.98	2.94–3.35	4.4
10^{-3}	1	17	45094	$1.73 \cdot 10^{-4}$	$5.891 \cdot 10^{-2}$	1.83	2.96–3.55	17.2

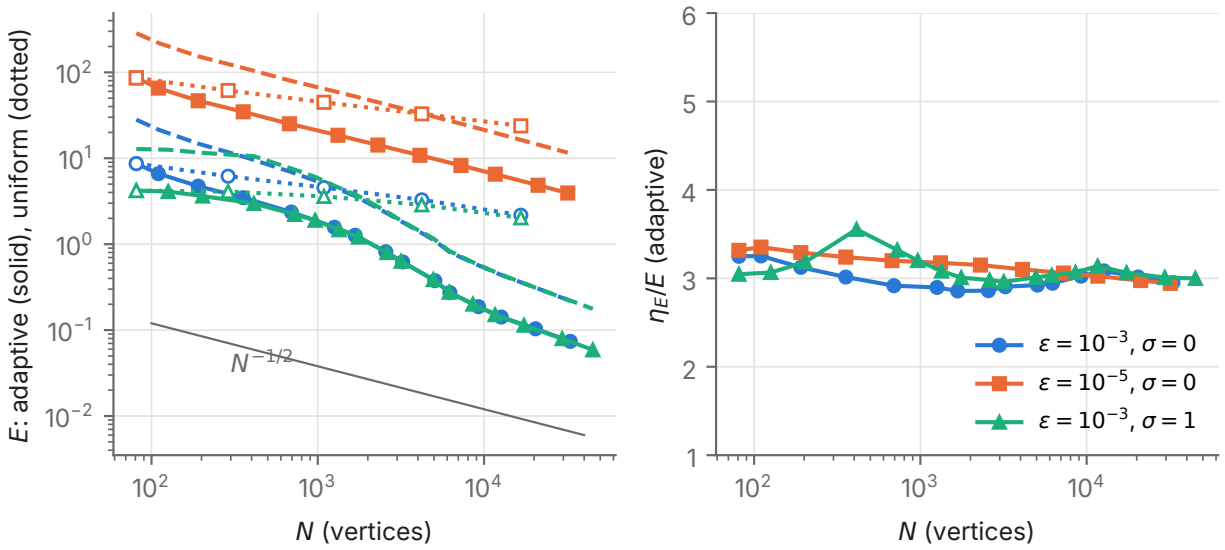


Figure 5: Adaptive refinement, layers problem, BJK. Left: E for adaptive (solid) and uniform (dotted, open markers) refinement, and η_E for the adaptive runs (dashed). Right: I_E along the adaptive runs.

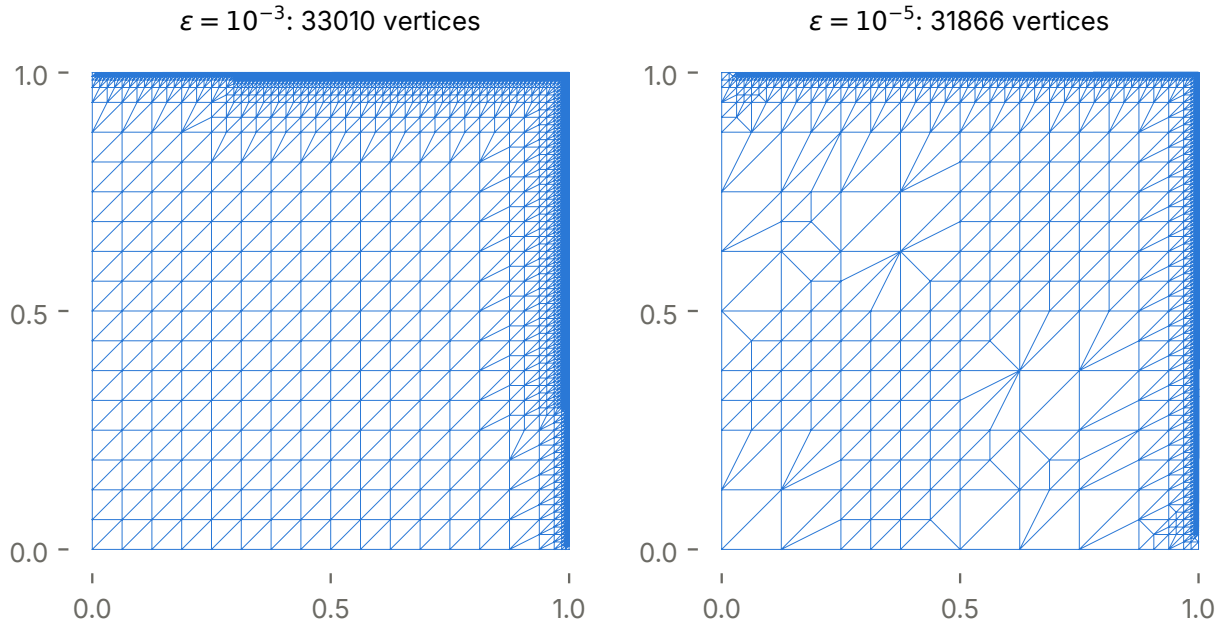


Figure 6: Final adaptive meshes, layers problem, $\sigma = 0$.

Refinement studies (Figure 7).

- **Effectivity.** Along the adaptive runs I_E lies in 2.83–4.18 and tends to the same values as on uniform meshes, about 3.0–3.2. The bound held in every loop.
- **Accuracy.** For the smooth solution, adaptive refinement gains nothing: at $N = 16\,641$ the two errors agree to within 1 %, and E converges at the rate 1 in both cases. For the layers problem the uniform error is 5.4 ($\varepsilon = 10^{-2}$) and 5.9 ($\varepsilon = 10^{-4}$) times the adaptive one at the same N. For $\varepsilon = 10^{-4}$, E converges at the rate 0.47 under uniform refinement and at 1.4–1.9 in the last adaptive loops, in which the layers become resolved.

Robustness sweep (Figures 8 and 9).

- $\sigma = 0$**.** At equal cost, $I_E = 2.89$–3.22 on the adaptive meshes against 2.80–3.23 on the uniform mesh, for all ε and both stabilisations, and it stays in 2.82–3.35 over all loops: the estimator is as robust on the meshes it generates as on the fixed mesh.
- $\sigma = 1$**, small** ε**.** The rise of I_E to 5.4–5.5 on the uniform mesh at $\varepsilon \le 10^{-7}$ (Section 10.4) does not occur at equal cost on the adaptive meshes, where $I_E = 2.75$–3.16. Figure 9 shows why. The large values, up to 9.6, appear only on the coarse initial mesh, and they come from the stabilisation part: $I_S = 17.8$ at $N = 81$ for $\varepsilon = 10^{-8}$. As the mesh resolves the layers, I_S falls to 1.5–1.6 and I_E returns to about 3. This confirms that the rise on the uniform mesh is a property of the under-resolved P_2 reference for $\|\tilde{S}\|_*$, not of the estimator.
- **Smooth problem.** For $\varepsilon \le 10^{-3}$ the adaptive indices are higher than the uniform ones at equal cost: $I_E = 4.0$–5.3 against 3.6–3.8 for $\sigma = 0$, and 4.5–6.0 against 3.9–6.0 for $\sigma = 1$. Along the runs they reach 7.8 after the first loop, and 10.9 on the initial mesh. For $\sigma = 1$ and $\varepsilon \le 10^{-6}$ the fixed-point iteration reaches the iteration limit $2 \cdot 10^4$ on most meshes beyond $N \approx 1100$, and the bound holds at these iterates.

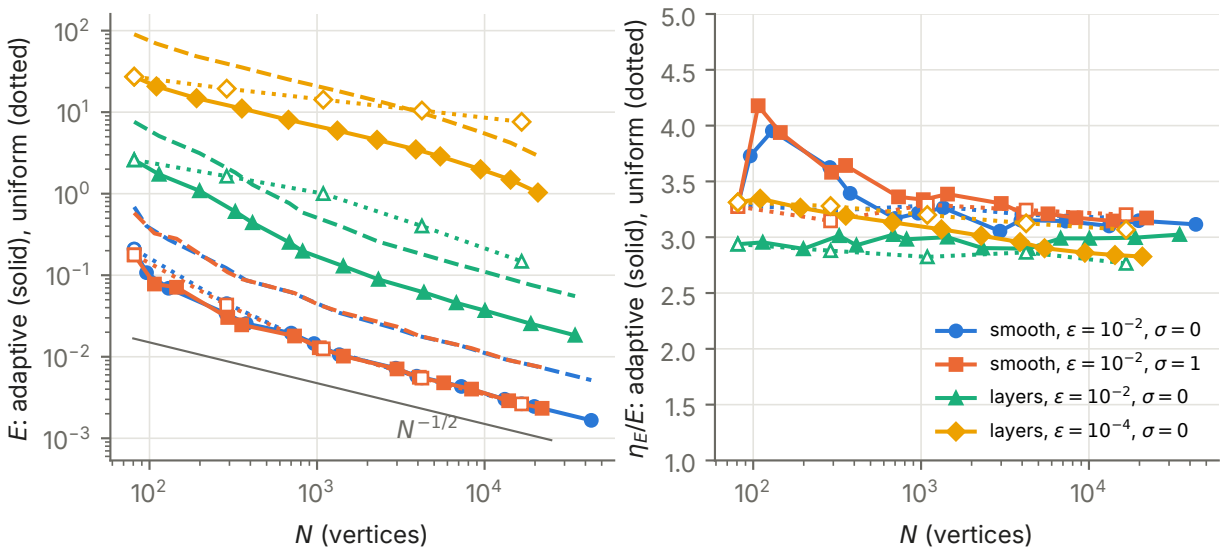


Figure 7: The cases of Table 1 under adaptive refinement. Left: E for adaptive (solid) and uniform (dotted, open markers) refinement, and η_E for the adaptive runs (dashed). Right: I_E, adaptive (solid) and uniform (dotted).

- **Energy norm.** At $\sigma = 0$ the energy-norm index still grows like ε^{-1} on the adaptive meshes, to $4 \cdot 10^5$ at $\varepsilon = 10^{-8}$ (uniform: $2.5 \cdot 10^6$); only the augmented measure is robust.
- **An open point.** For the layers problem with $\sigma = 1$ and $\varepsilon = 10^{-6}$, I_E rises again in the last loops, from 2.8 to 4.3, together with $I_\oplus$ (from 3.2 to 6.6). We have not identified the cause; it may be the convective part of the reference becoming resolved, and would need runs beyond $N = 10^4$.

Which refinement to use. For computing solutions, adaptive refinement is clearly preferable when the solution has layers, by factors 4–18 in E at equal N, and it is what the estimator is designed for. For testing the theory, the uniform meshes remain the primary evidence: a fixed coarse mesh keeps the layers unresolved, which is the regime in which robustness can fail, isolates the dependence on ε and σ, and is the only setting in which rates in h are meaningful. The adaptive runs confirm that the effectivity stays at about 3 on the meshes the estimator generates itself.

10.9 Inexact iterates and adaptive stopping

Iterates. Table 7 and Figure 10 evaluate the estimator at the fixed-point iterates u^i on the 16×16 mesh. The index I_E stays between 2.9 and 5.3 at every iterate, including the first one, where the algebraic part η_{alg} is up to 71 % of η_{disc}: this is part 2 of Proposition 8.1, efficiency at every iterate without any condition on the algebraic error. The ratio $\eta_{\text{alg}}/\eta_{\text{disc}}$ decreases monotonically for all problems except the smooth one with $\varepsilon = 10^{-6}$ and $\sigma = 0$, for which the fixed-point iteration does not converge in $2 \cdot 10^4$ iterations; there the ratio stalls between 10^{-3} and $4 \cdot 10^{-3}$. In that case the error is smallest at $i \approx 10$, $E(u^{10}) = 4.8$, and grows to 8.4 at $i = 2 \cdot 10^4$: further iterations make the iterate worse in E, and the estimator, which is efficient at every iterate, reflects this.

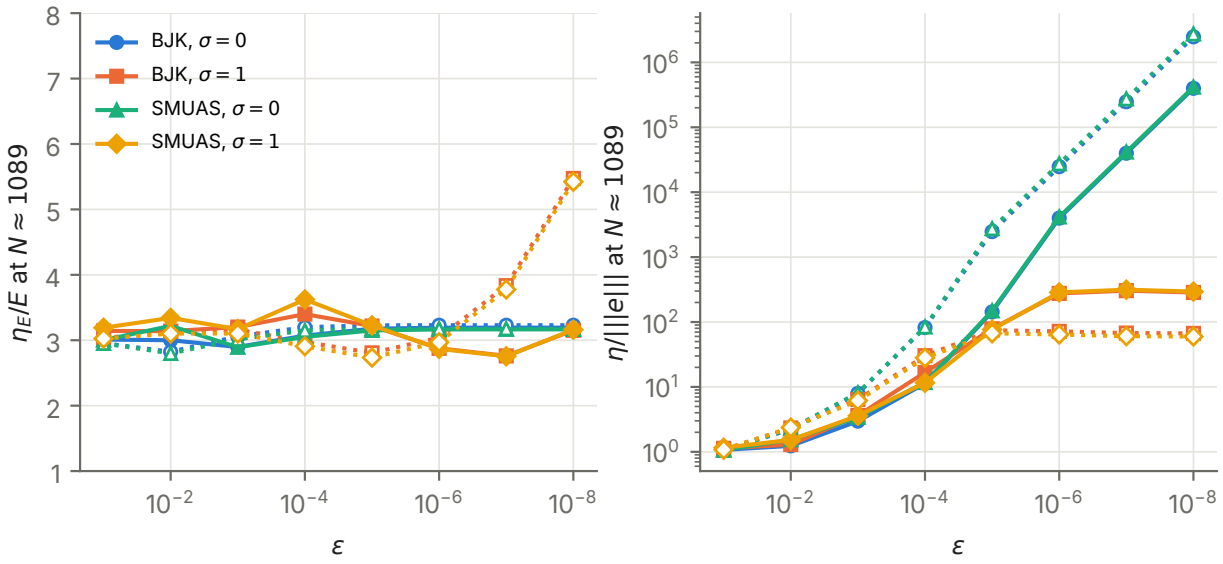


Figure 8: The robustness sweep of Table 2 at equal cost, layers problem: adaptive meshes with $N \approx 1089$ (solid) against the uniform 32×32 mesh (dotted, open markers). Left: I_E. Right: the energy-norm index $\eta/|||e|||$.

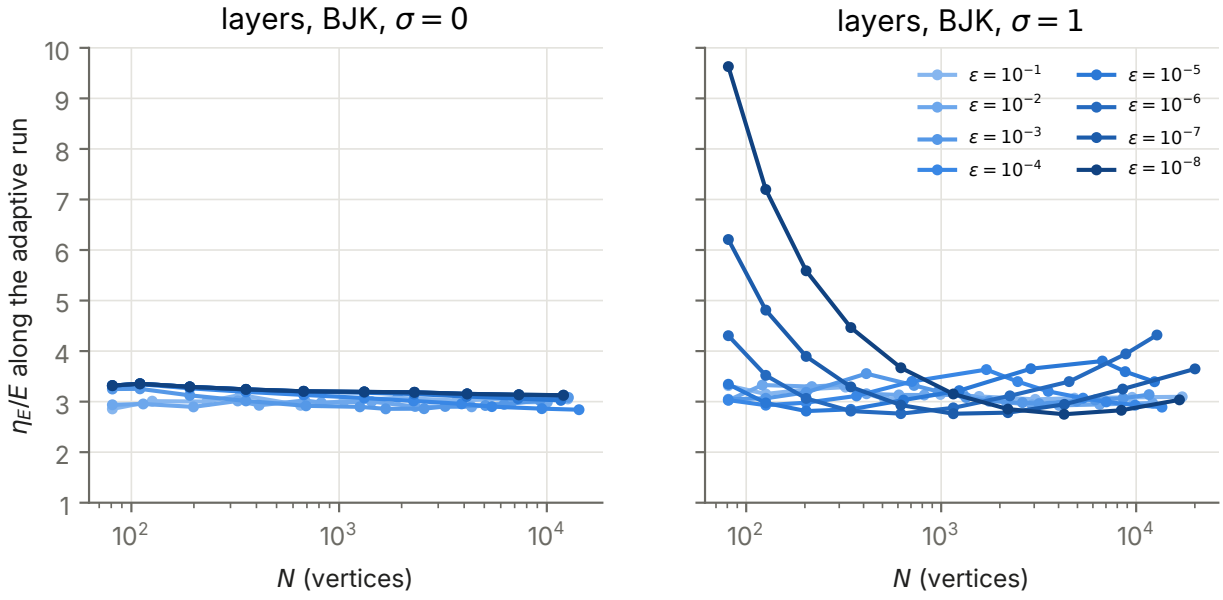


Figure 9: I_E along the adaptive runs of the robustness sweep, layers problem, BJK; one line per ε, darker for smaller ε.

Table 7: Fixed-point iterates u^i, $i \in \{1, 2, 5, \ldots, 1000\}$ below the converged count, and the converged solution; BJK, $n = 16$. The split is $\eta_{\text{disc}} = \eta_E - \eta_{\text{alg}}$.

problem	ε	σ	it. to tol.	η_E/E over i	$\max_i \eta_{\text{alg}}/\eta_{\text{disc}}$	$E(u^1)/E(u^{\text{last}})$	$E \le \eta_E$
smooth	10^{-2}	0	39	3.24–3.63	0.71	1.81	yes
smooth	10^{-2}	1	37	3.15–3.45	0.51	1.58	yes
smooth	10^{-6}	0	20000	3.76–4.61	0.58	1.06	yes
smooth	10^{-6}	1	664	4.42–5.28	0.19	1.20	yes
layers	10^{-2}	0	47	2.88–2.97	0.02	1.01	yes
layers	10^{-2}	1	45	3.32–3.33	0.01	1.01	yes
layers	10^{-6}	0	635	3.29–3.47	0.04	1.01	yes
layers	10^{-6}	1	502	3.51–3.56	0.02	1.03	yes

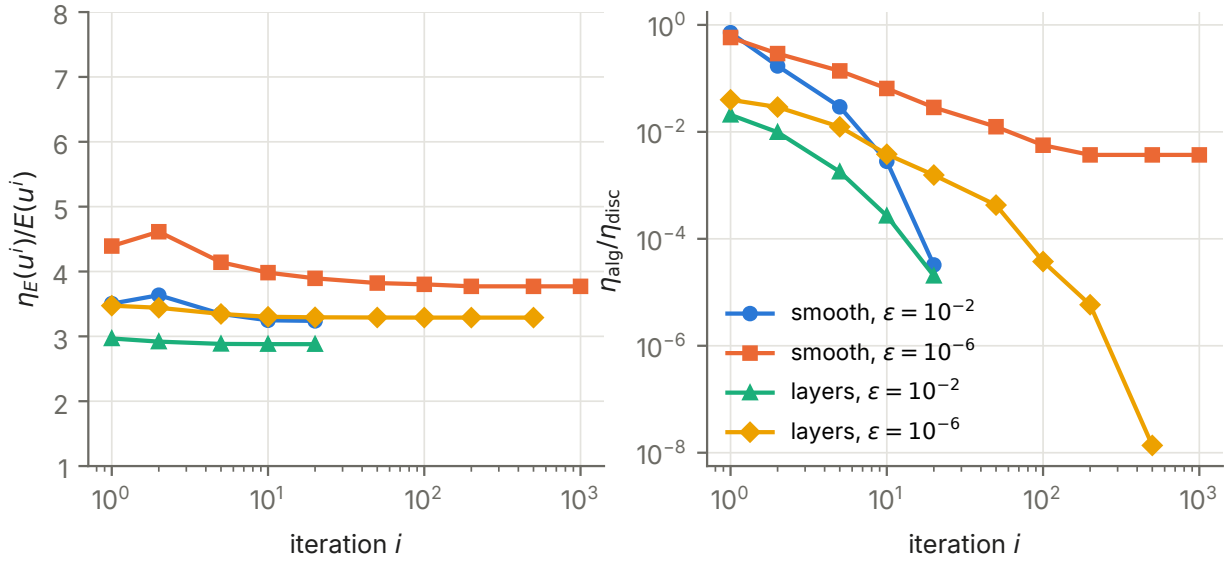


Figure 10: Fixed-point iterates, BJK, $\sigma = 0$, $n = 16$. Left: I_E at the iterate u^i. Right: the ratio $\eta_{\text{alg}}/\eta_{\text{disc}}$ (points with a ratio below 10^{-8} are omitted).

Adaptive stopping. Table 8 runs five adaptive loops as in Section 10.7 in which the fixed-point iteration is stopped either at the fixed tolerance $\|r\| \le 10^{-10}\sqrt{N}$ or by the criterion (8.1), $\eta_{\mathrm{alg}} \le \gamma\eta_{\mathrm{disc}}$, with $\eta_{\mathrm{disc}} = \eta_E - \eta_{\mathrm{alg}}$; each loop starts from the zero initial guess.

- **Savings.** With $\gamma = 0.1$, the total number of iterations drops from 227 to 25 (smooth, $\varepsilon = 10^{-2}$) and from 2323 to 25 (layers, $\varepsilon = 10^{-4}$), by factors 9 and 93; with $\gamma = 0.01$ the factors are 4.5 and 46.
- **Accuracy.** The final error E changes by at most 6 % ($\gamma = 0.1$, layers) and by at most 0.3 % for $\gamma \le 0.01$. The index I_E stays in 3.1–4.8 in all runs, and in every stopped loop $\eta_{\mathrm{alg}}/\eta_{\mathrm{disc}} \le \gamma$, so $E \le (1+\gamma)\eta_{\mathrm{disc}}$ (Proposition 8.1, part 3).
- **Non-converging iteration.** For the smooth problem with $\varepsilon = 10^{-6}$ the fixed-point iteration does not reach the tolerance within 10^4 iterations on any mesh, and 50 000 iterations give $E = 3.22$. The criterion with $\gamma = 0.1$ stops after 5–15 iterations per loop, 50 in total, with a *smaller* error, $E = 1.98$: as in Figure 10, the early iterates are better than the late ones. For $\gamma = 10^{-3}$ the criterion fires in only one of the five loops, since $\eta_{\mathrm{alg}}/\eta_{\mathrm{disc}}$ stalls at about $3 \cdot 10^{-3}$; this is a property of the iteration, not of the estimator.

Table 8: Adaptive stopping in five adaptive loops (dual marking by the SV indicators, red–green refinement), BJK, $\sigma = 0$. "tol." is the fixed tolerance $\|r\| \le 10^{-10}\sqrt{N}$; otherwise the iteration stops at $\eta_{\mathrm{alg}} \le \gamma\eta_{\mathrm{disc}}$ (at most 10^4 iterations). The last column says whether every loop stopped before the iteration limit.

problem	ε	γ	it./loop	total	E (last)	η_E/E	max $\eta_{\mathrm{alg}}/\eta_{\mathrm{disc}}$	stopped
smooth	10^{-2}	tol.	30–59	227	$2.550 \cdot 10^{-2}$	3.29–3.95	$5.78 \cdot 10^{-8}$	yes
smooth	10^{-2}	0.1	5	25	$2.554 \cdot 10^{-2}$	3.41–3.97	$5.07 \cdot 10^{-2}$	yes
smooth	10^{-2}	0.01	10	50	$2.551 \cdot 10^{-2}$	3.33–3.98	$7.27 \cdot 10^{-3}$	yes
smooth	10^{-2}	0.001	10–20	80	$2.551 \cdot 10^{-2}$	3.29–3.96	$9.89 \cdot 10^{-4}$	yes
smooth	10^{-6}	tol.	10000	50000	3.218	3.77–4.58	$3.03 \cdot 10^{-3}$	no
smooth	10^{-6}	0.1	5–15	50	1.984	4.04–4.79	$9.83 \cdot 10^{-2}$	yes
smooth	10^{-6}	0.01	120–760	2190	2.749	3.80–4.45	$9.10 \cdot 10^{-3}$	yes
smooth	10^{-6}	0.001	7320–10000	47320	3.218	3.77–4.58	$3.03 \cdot 10^{-3}$	no
layers	10^{-4}	tol.	385–628	2323	8.044	3.13–3.34	$3.57 \cdot 10^{-9}$	yes
layers	10^{-4}	0.1	5	25	8.524	3.20–3.42	$2.49 \cdot 10^{-2}$	yes
layers	10^{-4}	0.01	10	50	8.023	3.15–3.38	$8.43 \cdot 10^{-3}$	yes
layers	10^{-4}	0.001	25–50	175	8.04	3.14–3.35	$9.29 \cdot 10^{-4}$	yes

10.10 Variable diffusion

Table 9 reports the variable-diffusion problems on the meshes $n = 8, 16, 32$. All bounds are guaranteed. For the smooth coefficient, the oscillation η_ε converges at the rate 2.09–2.13 in h, in accordance with Lemma 9.1(iii), and is at most 12 % of η (at contrast 10^4 on the coarsest mesh). For the jump aligned with the mesh, $\Pi_1\varepsilon = \varepsilon$ and $\eta_\varepsilon = 0$. The effectivity I_E lies in 2.6–4.5 for contrasts up to 10^4, for $\varepsilon = 10^{-2}$ and 10^{-6} alike. For the jump it does not grow with the contrast; for the smooth coefficient it grows mildly, from 3.1 at contrast 1 to at most 4.3 at contrast 10^4 (on the coarsest mesh). This is the quasi-monotone case; the constants of Theorem 9.2 depend on the patch contrast, but for an aligned jump that contrast does not show up in the indices. Checkerboard (non-quasi-monotone) coefficients are not tested; for them no local estimator can be robust [Li26].

Table 9: Variable diffusion, BJK, $\sigma = 0$, $n \in \{8, 16, 32\}$: ranges over the three meshes. The rate is that of η_ε with respect to $N^{-1/2}$ between $n = 16$ and $n = 32$ (expected: 2).

$\varepsilon(x)$	ε	contrast	I	$I_\oplus$	I_E	$\max \eta_\varepsilon/\eta$	rate	$E \le \eta_E$
smooth	10^{-2}	1	1.23–4.58	3.30–4.06	3.07–3.13	0.000	—	yes
smooth	10^{-2}	100	1.07–1.24	3.16–3.39	3.15–3.26	0.035	2.09	yes
smooth	10^{-2}	10000	1.10–1.44	3.30–4.28	3.29–4.27	0.117	2.09	yes
smooth	10^{-6}	1	7959–36796	3.72–5.01	3.50–3.70	0.000	—	yes
smooth	10^{-6}	100	818–3354	3.73–5.44	3.54–4.03	0.000	2.13	yes
smooth	10^{-6}	10000	64.07–336	4.31–5.49	3.94–4.46	0.001	2.10	yes
jump	10^{-2}	100	2.56–7.63	3.43–4.18	2.61–2.99	0.000	—	yes
jump	10^{-2}	10000	2.61–8.12	3.43–4.22	2.60–3.01	0.000	—	yes
jump	10^{-6}	100	9025–28125	4.75–5.12	3.44–3.54	0.000	—	yes
jump	10^{-6}	10000	2156–4989	4.78–5.08	3.41–3.52	0.000	—	yes

10.11 Three dimensions

Table 10 repeats the uniform-refinement study on the unit cube ($n = 4, 8, 16$, up to $N = 4913$ vertices and 24 576 tetrahedra).

- **Layers.** $I_E = 2.94$–3.55 for $\varepsilon = 10^{-2}$ and 10^{-6}, as in two dimensions; $I_\oplus \le 4.9$ and $I_S = 1.21$–1.26.
- **Smooth solution.** I_E is higher than in two dimensions, 5.1–6.4, but decreases under refinement for both values of ε ($5.97 \to 5.40$ and $6.39 \to 5.09$). These meshes are coarse: E still converges at the pre-asymptotic rates 2.1–2.7 in h, dominated by the stabilisation part $\|\tilde{S}\|_*$, which is estimated sharply ($I_S = 1.28$–1.31); the larger index comes from the convective dual norm on coarse tetrahedral meshes. For $\varepsilon = 10^{-6}$ on the finest mesh the fixed-point iteration does not converge within $2 \cdot 10^4$ iterations, and the bound still holds at the last iterate.
- **Energy norm.** As in two dimensions, I is useless for $\sigma = 0$ and small ε ($1.0 \cdot 10^4$–$1.4 \cdot 10^5$ at $\varepsilon = 10^{-6}$, against 2.6–9.3 at $\varepsilon = 10^{-2}$), while I_E is not affected.
- **Cost.** The estimator takes about 8 s at $N = 729$ and 220–240 s at $N = 4913$ in our Python implementation; the cost is dominated by the vertex-patch problems in RTN_1, whose size grows quickly with the number of tetrahedra per patch in three dimensions. The patch problems are independent and can be solved in parallel; the global RT_0 defect problem takes a small fraction of the time.

10.12 Summary

The experiments answer the questions (Q1)–(Q7) as follows.

(Q1) The bound $E \le \eta_E$ held in every run: two and three dimensions, both stabilisations, $\varepsilon = 10^{-1}, \dots, 10^{-8}$, $\sigma \in \{0, 1\}$, variable reaction and diffusion, adaptive meshes and inexact iterates, and so did $I \ge 1$, $I_\oplus \ge 1$ and $I_S \ge 1$. Since the references are lower bounds of the dual norms, this is a consistency check rather than a proof.

(Q2) For constant coefficients, I_E lies in 2.6–5.3 in two dimensions, inexact iterates included, independently of ε, σ, the stabilisation and the mesh; the only exception is a rise to 6.0 ($n = 32$) and 7.1 ($n = 16$) at $\sigma = 1$, $\varepsilon \le 10^{-7}$, which the adaptive runs identify as an under-resolved reference for $\|\tilde{S}\|_*$ (Section 10.8), as the asymptotically exact reference (10.1) confirms ($I_S = 1.6$–2.2 and $\eta_E/E = 8.3$–10.2 for θ down to 10^{-5}); in three dimensions it is 2.9–6.4 and decreases

Table 10: Three dimensions, uniform refinement of the unit cube, BJK, $\sigma = 0$. Rates with respect to $N^{-1/3} \sim h$; the last column is the time of the estimator in seconds.

problem	ε	N	E	rate	I	$I_\oplus$	I_E	time
smooth	10^{-2}	125	0.7		9.27	7.82	5.97	0.7
smooth	10^{-2}	729	0.144	2.69	4.83	6.98	5.71	9.0
smooth	10^{-2}	4913	$3.55 \cdot 10^{-2}$	2.20	2.64	5.84	5.40	226.6
smooth	10^{-6}	125	$6.72 \cdot 10^{1}$		$7.4 \cdot 10^{4}$	9.26	6.39	0.6
smooth	10^{-6}	729	$2.00 \cdot 10^{1}$	2.06	$2.8 \cdot 10^{4}$	6.93	5.31	8.4
smooth	10^{-6}	4913	4.06	2.51	$1.0 \cdot 10^{4}$	5.56	5.09	238.0
layers	10^{-2}	125	2.74		6.67	4.01	3.10	0.6
layers	10^{-2}	729	1.82	0.70	4.31	3.71	3.01	8.0
layers	10^{-2}	4913	1.14	0.73	2.93	3.64	2.94	217.3
layers	10^{-6}	125	$2.79 \cdot 10^{2}$		$1.4 \cdot 10^{5}$	4.71	3.38	0.5
layers	10^{-6}	729	$2.01 \cdot 10^{2}$	0.56	$7.3 \cdot 10^{4}$	4.78	3.47	8.5
layers	10^{-6}	4913	$1.35 \cdot 10^{2}$	0.63	$3.8 \cdot 10^{4}$	4.93	3.55	236.1

under refinement. The energy-norm index I, and the guaranteed bound of [ABR17], are not robust.

(Q3) The effectivity grows with c_b, at most linearly for $\eta_{E,c}$ and well below c_b^2 for η_E: a variable reaction with a small lower bound is the one case in which robustness is lost, as the theory predicts.

(Q4) The local ratio $\max_K \eta_K^2 / E_{\mathrm{loc}}(\tilde{\omega}_K)^2$ is at most 0.51.

(Q5) Adaptive refinement driven by η_K resolves the layers and reduces E by factors 4–18 against uniform refinement at equal N, with $I_E = 2.9$–3.6 throughout. Repeating the refinement and robustness studies adaptively gives the same effectivity, about 3, as on uniform meshes.

(Q6) The estimator is efficient at every iterate, and the stopping criterion saves up to 99 % of the iterations at a change of the error of at most 6 % ($\gamma = 0.1$) or 0.3 % ($\gamma \le 0.01$).

(Q7) For variable diffusion, η_ε converges at rate 2 and I_E stays in 2.6–4.5 up to contrast 10^4.

Reproducibility. All results are produced by the scripts `run_numerics.py` and `make_figures.py` in the directory `examples/convection_diffusion/sv_numerics` of `galerkit`. The first runs the experiments (about 35 minutes on four cores, the three-dimensional runs sequentially because of their memory use); the second writes the tables, the figures and, for every stored solution, a VTK file (`.vtu`) with u_h, u, $u - u_h$, ε, η_K, $|||e|||_K$ and the local effectivity $\eta_K / |||e|||_K$. The figures of Section 10.8 are produced by `make_adaptive_figures.py` in the same directory.

11 Conclusions and open problems

We have shown that the Smears–Vohralík-type equilibrated-flux estimator, extended to convection with the stabilisation as an additional diffusion tensor, is guaranteed and robustly locally efficient for algebraically stabilised convection–diffusion–reaction problems. The proof needs no weights in the patch problems and no use of the discrete equation, so it covers inexact iterates and yields a stopping criterion without smallness conditions. Several questions remain open.

- **Removing c_b.** The factors $(1 + c_b)^2$ and $1 + c_b$, the restriction of Theorem 7.2 to $\sigma > 0$ or $c = \frac{1}{2} \operatorname{div} b$, and the additional term of Remark 7.3 all stem from measuring the reaction by

its lower bound σ. An energy norm with the full effective reaction $c - \frac{1}{2}\operatorname{div} b$, with cellwise weights in the spirit of [AV14], would remove all three.

- **Non-P_1 Dirichlet data.** The efficiency of the Dirichlet lifting terms is not treated.
- **Robustness in the diffusion contrast** is shown only numerically, and only for quasi-monotone ε.
- **The stopped iterate against the discrete solution**, $E(u^i) \lesssim E(u_h)$, needs a discrete stability estimate for the nonlinear operator.
- **The direct variant** (Remark 3.2) has no efficiency result.

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